Panasonic

Ver. 2018.0910

Specification

【 Electronic Component Mounting System 】

MES for Electronics Assembly

Model ID : PanaCIM-EE

Model ID : NM-EJS3B

Panasonic Smart Factory Solutions Co., Ltd.

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12. The contents of this document are valid as of September 10, 2018.

⚠ Precautions for Safe Use	
● Thoroughly read the instruction manual and use the machine only as directed.	**● The machine described in this Specification is an industrial machine. Confirm the operation requirements described in the separate instruction manuals. Regarding machine installation, operation and maintenance, and handling of necessary materials, the local laws and requirements may be applied.** **● Before carrying out the operation and maintenance of this machine, confirm the descriptions in the separate instruction manuals and warning information attached on this machine regardless of the operating condition of this machine. Ignoring safety functions might cause injury, electric shock or machine trouble.**

Revision History

This document content is based on the version of the last date printed below.

Revision date	Version	Revised page	Revisions
Oct. 15, 2012	Ver. 2012.1015	---	First edition
Apr. 18, 2013	Ver. 2013.0418	P. 31, 32	Addition of Applicable Model: NPM-D2
Jun. 3, 2013	Ver. 2013.0603	P. 3, 4, 8-13, 16, 23	・Addition of Express package ・Addition of applicable handy scanner: MC55
Sep. 5, 2013	Ver. 2013.0905	P. 3, 5, 6, 9, 13, 14, 16, 19, 20, 30, 34, 35	Addition of Applicable Model: AM100, SPG
Nov. 1, 2013	Ver. 2013.1101	P. 3, 14	Addition of option for Screen printer main body: PanaCIM-EE ready (License)
May 12, 2014	Ver. 2014.0512	P. 4, 6, 11, 13, 14, 16, 20, 30, 31, 34 ~ 36	・Addition of Applicable Model: NPM-D3 ・Addition of Production Monitoring client
Jun. 2, 2014	Ver. 2014.0602	P. 6, 9, 16	・Multiple NPM-DGS Connection to one PanaCIM ・PanaCIM Express (Material verification): AM-LNB support
Dec. 10, 2014	Ver. 2014.1210	P. 13, 34, 35	・Addition of Applicable Model: NPM-W2
Mar. 2, 2015	Ver. 2015.0302	P. 4 ~ 8, 11, 12, 16 ~ 18, 20, 23, 24, 35, 36	・Revision of system equipment configuration ・Revision of hardware requirements
Jul. 24, 2015	Ver. 2015.0724	P. 1 ~ 6, 8, 10 ~ 14, 16, 25, 31 ~ 39	・Addition of Component Supply Navigator(others)
Jun. 17, 2016	Ver. 2016.0617	P. 1, 3 ~ 6, 8 ~ 11, 13, 14, 16, 19, 30, 36, 37, 39 ~ 41	・Addition of support for NPM-TT2, NPM-DX, and SPV ・Addition of serverless verification package ・Upgraded from CD to DVD ・Revision of supported Internet Explorer version
Oct. 20, 2016	Ver. 2016.1020	P. 1 ~ 3, 5 ~ 9, 11 ~ 19, 34, 41, 42, 45, 49	・Revision of system equipment configuration ・Addition of Maintenance module ・Revision of hardware requirements
Mar. 12, 2018	Ver. 2018.0312	P. 5 ~ 6, 11, 15 ~ 18, 20 ~ 21, 46 ~ 52	・Update of network configuration ・Change of capable Model No. of access point, etc. ・Revision of maintenance server's equipment configuration and hardware specification ・Revision of Model No. for handy scanner: ・Revision of supported model IDs of Component Supply Navigator ・Addition of Model IDs: NPM-WX, NPM-WXS, NPM-W2S ・Addition of supported OS and SQL Server
Sep. 10, 2018	Ver. 2018.0910	P. 57	・Addition of "Important Notes"

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1. General Description

PanaCIM-EE (Enterprise edition) is support software designed to assist you in visualizing / bringing efficiency to / ensuring quality of your operations and production in the electronics assembly process, as well as providing you with MES*2 functions through the use of its 9 constituent modules*1 and thereby enhancing OEE*3.

■ **Module**

1	Material Verification*4	Verifies packaging and assembly materials, and electronic components
2	Production Analysis*4	Collects / analyzes production performance
3	Traceability*4	Prepares / controls PCB- or placement point-level trace information
4	Material Control*4	Controls materials / parts on the assembly floor
5	Production Changeover & Control*4	Controls automatic changeover of the product handled by the machine through the use of PCB IDs, etc.
6	Maintenance	Manages maintenance of equipment used in the assembly process, such as the machine and the feeder
7	Production Monitor & Dispatch*4	Actualizes the visualization of the production status of the machine, and others.
	Component Supply Navigator Option	Optimizes the supply sequence to provide a shortest route without component shortage.
8	Enterprise Link*4	Interface for your high-order system

*1 A module refers to a cluster of software application functions, such as Material Verification, which is provided in units of the major operation or classification in the assembly process.
*2 MES: Manufacturing Execution System
*3 OEE: Overall Equipment Effectiveness
*4 Any combination of your choice is acceptable.
The system (module) configuration available and the number of pieces of target equipment depend on your license type.

2. Features

Material Verification

A support function, performed in the case of setup for production / product changeover / splicing / parts shortages, which compares (verifies) material or parts information to (and) production programs before placement of materials or parts and thereby prevents the wrong materials or parts from being placed in the machine.

The serverless verification package provides the function for verifying parts line by line (equivalent to the LNB parts verification) that does not require PanaCIM server PC. Refer to “6. Serverless Verification Package” for details.

Traceability

A support function, designed to help control quality and meet market requirements, which controls history information on fabricated PCBs through the combined use of information such as the production performance record of the machine or material / parts information, and the PCB information read by the barcode reader on the machine or the conveyor during the assembly process.

Production Analysis

A support function that, for the purpose of enhancing productivity (improved recognition / pick-up error rate, reduced downtime, etc.) in the assembly line, collects operating information of the machine and then analyzes / provides statistics of / displays in real time the information.

Material Control

A support function used to manage the number and location of parts or feeders in inventory in the production or preparation area and also to reserve parts. The moisture control function exercises expiration date control over moisture sensitive devices (MSD).

Production Changeover & Control

A support function intended to reduce product changeover time through automatically downloading a production program and also automatically adjusting the lane width according to the PCB ID.

Production Monitor & Dispatch

A support function that obtains production / operating information in real time and displays the condition of the line on the monitor or PDA, which provides to the operator instructions on what to do in the event of an error or parts shortages and thus helps enhance productivity, thereby reducing the downtime of the assembly line or allowing for advance preparation of parts.
The Component Supply Navigator (option) optimizes the supply sequence to provide a shortest route without component shortage. It can prevent short time breakdown in the production line and contribute to manpower saving.

Enterprise Link

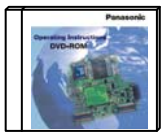
A support function designed to output to external applications the data (material verification data, trace data, operating performance data) collected from the machine during production.

Maintenance

A support function used to manage the profile of the assets (machines, heads, feeders, etc.) on the assembly floor and to record the history of maintenance per asset.
This helps reduce the time required for maintenance planning and management operations.

3. Specification

3.1 Constituent Devices

<table>
<tr><th></th><th>Item</th><th colspan="2">Description</th></tr>
<tr><td>1</td><td>PanaCIM-EE software</td><td colspan="2"> 
Installation disk
(for Windows & Linux)
 
Manual
(DVD-ROM*)
 
USB dongle
 
License disk
(Based on the number of selected modules)
* The manual (bound version) is optional.</td></tr>
<tr><td>2</td><td>Wireless scanner and related equipment

* These are prepared by sales companies, agents, or customers.</td><td colspan="2"> 
Wireless scanner*1
 
Cradle*2
 
Spare battery
 
Access point*3
 
Hub *4
 
LMNC
 
Android tablet</td></tr>
<tr><td>3</td><td>PC and electric source

* These are prepared by sales companies, agents, or customers.</td><td colspan="2"> 
Power cable and power strip
 
PC / Server for PanaCIM-EE
 
LAN cable*5
 
UPS (Uninterruptible Power Supply)*6</td></tr>
<tr><td rowspan="5">4</td><td rowspan="5">Machine side options

* Select from the options on each equipment piece.</td><td>NPM X/ NPM / AM series</td><td>PanaCIM-EE ready (License)</td></tr>
<tr><td>Support station*7</td><td>Support station box (With wired scanner*8)</td></tr>
<tr><td>CM series main body</td><td>PanaPRO/CVT supporting unit: For MC70/ MC75*9</td></tr>
<tr><td>Change Cart Preparation Unit*7</td><td>PanaPRO/CVT supporting unit: For MC70/ MC75*9</td></tr>
<tr><td>Screen printers</td><td>PanaCIM-EE ready (License)</td></tr>
</table>

*1 Designate MC55 as the wireless scanner.
*2 Comes in sets with a cradle body, AC adapter, and power and communication cables.
*3 Comes in sets with an access point body, 3 antennas, a power injector, power cable and the manual (DVD).
The initial setup requires RS-232C cross cable (D-sub 9-pin with female connector on both ends).
*4 Must be 1000 BASE-T-ready, and requires more than 5 systems.
*5 Cables must be routed between PC for PanaCIM-EE and hub / Access point and hub / Power injector and Access point / PT200 and Hub / Change cart preparation unit and hub / (the machine and hub). Specifications should be Category 5 or higher.
*6 Contact the maker of the server you have selected before selecting an appropriate device.
*7 Required for off-line setup.
*8 The wireless scanner (MC55) is used for material verification.
*9 Select “For MC70/75” for the optional specifications for the machine “PanaPRO/CVT supporting unit.”
“Cradle holder” and “Power supply” of the wireless scanner will be added.
(Only “Cradle holder” for the change cart preparation unit.)

3.2 System Configuration

System equipment configuration

Constituent elements	OS / Database	Role
	Restrictions	
PanaCIM application server*4	Windows Server® 2008 R2 (64-bit version)*2, 3 Windows Server® 2012 R2 (64-bit version)*2, 3 Windows Server® 2016 (64-bit version)*2, 3 Microsoft® SQL Server® 2008 R2 Standard/ Enterprise Edition (32-bit/ 64-bit version)*2, 3, 5 Microsoft® SQL Server® 2012 Standard/ Enterprise Edition (32-bit/ 64-bit version)*2, 3 Microsoft® SQL Server® 2016 Standard Edition (64-bit version)*2, 3, 5	・Application host ・Provision of Web portal (Web server) ・License management ・Data control (Database) ・The minimum configuration RAID 1 is preferred
	・Internet Explorer® 11 is required. *For IE8, turn ON “Compatibility View”.	
PanaCIM controller server*4	Red Hat Enterprise Linux 5.2/5.4/5.7/6.5/6.7 Server (32-bit version)*3	・Server that controls line configuration, various settings and the data, events or the like of all PanaCIM applications ・The minimum configuration RAID 1 is preferred
	・Up to 100 stages are controllable only with the PanaCIM controller server	
PanaCIM maintenance server*4	Windows Server® 2012 R2 (64-bit version)*2, 3, 5 Windows Server® 2016 (64-bit version)*2, 3, 5 Microsoft® SQL Server® 2012 Standard/Enterprise Edition (32-bit/ 64-bit version)*2, 3 Microsoft® SQL Server® 2016 Standard Edition (64-bit version)*2, 3	・Asset profile information management (Database) ・Web server for Android tablet ・Maintenance unit communication interface ・The minimum configuration RAID 1 is preferred
	・Google Chrome™ or Internet Explorer® 11 is required.※6	
PT interface	Windows® 7 Professional (32-bit/ 64-bit version) Windows® 8.1 Professional (64-bit version)	・Carries out data communication with PT200
	・A single unit required per PT200	
Web client (Report reference PC)	Windows® 7 Professional (32-bit/ 64-bit version) Windows® 8.1 Professional (64-bit version) Windows® 10 Pro (64-bit version)	・Accesses the PanaCIM application server and produces graph / report displays ・Accesses the PanaCIM maintenance server and produces graph / report displays
	・A new purchase is not required. It is available with your existing PC that allows access to the PanaCIM application server. (The response speed and others depend on your network environment) ・You do not have to use the PanaCIM application server to access reports. ・Internet Explorer® 11 is required. *For IE8, turn ON “Compatibility View”.	
PanaCIM Production Monitor & Dispatch client	Windows® 7 Professional (32-bit/ 64-bit version) Windows® 8.1 Professional (64-bit version)	・Carries out data communication for Production Monitoring
	・Required if the number of stages is more than 7 per line and the number of lines is more than 4.*1	
LMNC*1	—	・Communication with the installed barcode reader ・PCB barcode checking ・Control of the loading of PCBs into the placement machine
	・Required for Traceability and Production Changeover & Control (CM-/SP-based machines only)	

Constituent elements	OS / Database	Role
	Restrictions	
Installed barcode reader*1	—	・Reads the barcode of a PCB being loaded into the machine
	・Required for Traceability and Production Changeover & Control	
Access point*1	—	・Wireless communication between the PanaCIM controller server and the scanner
	・Required for Material Verification, Material Control and Production Monitor & Dispatch	
Wireless scanner*1	—	・Reads barcodes while in the Material Verification or Material Control process ・Displays events such as parts shortages during Production Monitor & Dispatch
	・Required for Material Verification, Material Control and Production Monitor & Dispatch	
Android tablet*1	Android™ 6.0	・Displays the check items for maintenance ・Displays the spare parts information
	・Google Chrome™ is required.*1 ・Android tablet and Google Chrome™ are required for the Maintenance module to run.	

*1 Equipment prepared by you, which is required according to the module (function) you use.
The necessary number of pieces of equipment depends on the machine configuration. So, for details, please contact us.

*2 As for Windows Server® and Microsoft® SQL Server®, you need to have licenses to use the servers and also permissions to have access to the servers. Please contact Microsoft® or a sales agency, and prepare necessary licenses or permissions.
Note that it should be unified under one standard, either 32-bit or 64-bit version.

*3 If the supported versions of these OS or database are hardly available, purchase the applicable software products from vendors. In doing so, make sure that downgrade rights are granted.

*4 In case of a single server configuration, PanaCIM application server, PanaCIM control server and PanaCIM maintenance server can run on a single server.

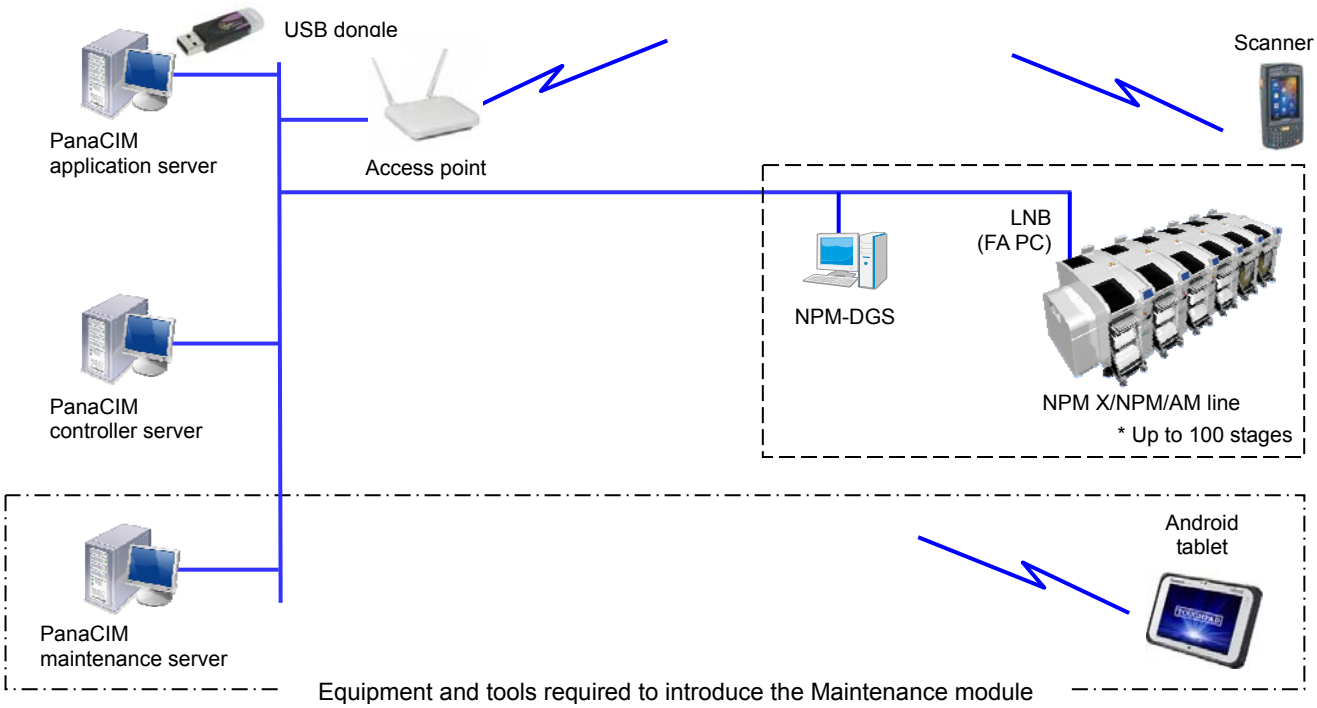
*5 In newly purchasing Windows Server after October 1, 2016, Windows Server 2012R2 is no longer available because Windows Server 2016 has started to be sold.
Note the followings below:
1） License selling method has been changed since Windows Server 2016.
One physical server requires at least 16 core licenses.
（Minimum purchase unit：At least eight core licenses are required for one physical processor, and besides those for at least two processors are required）

*6 We prefer HTML5-supported Google ChromeTM.

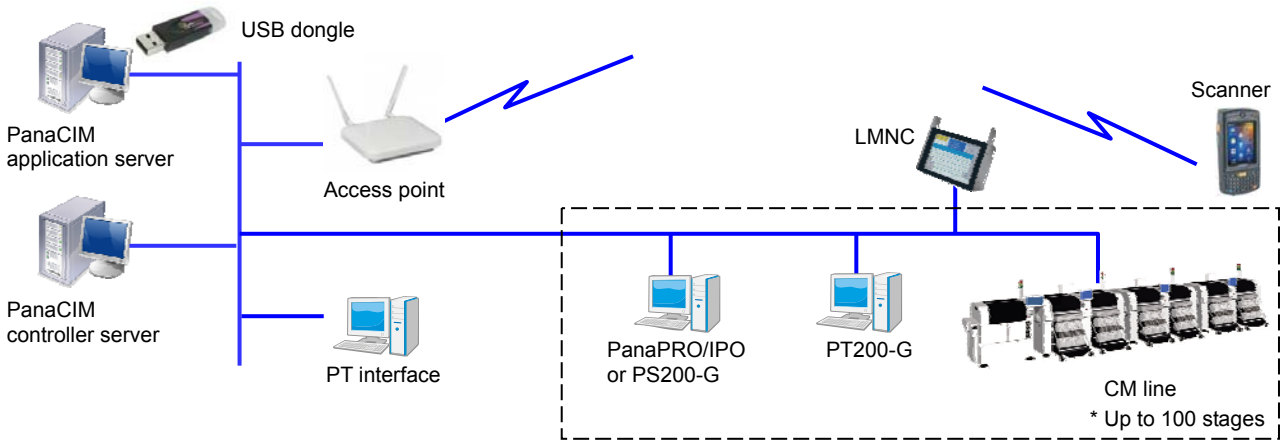
Small-scale configurations of PanaCIM

Up to a maximum of 100 stages can be controlled with just a single PanaCIM.

For NPM X/NPM/AM line



For CM line

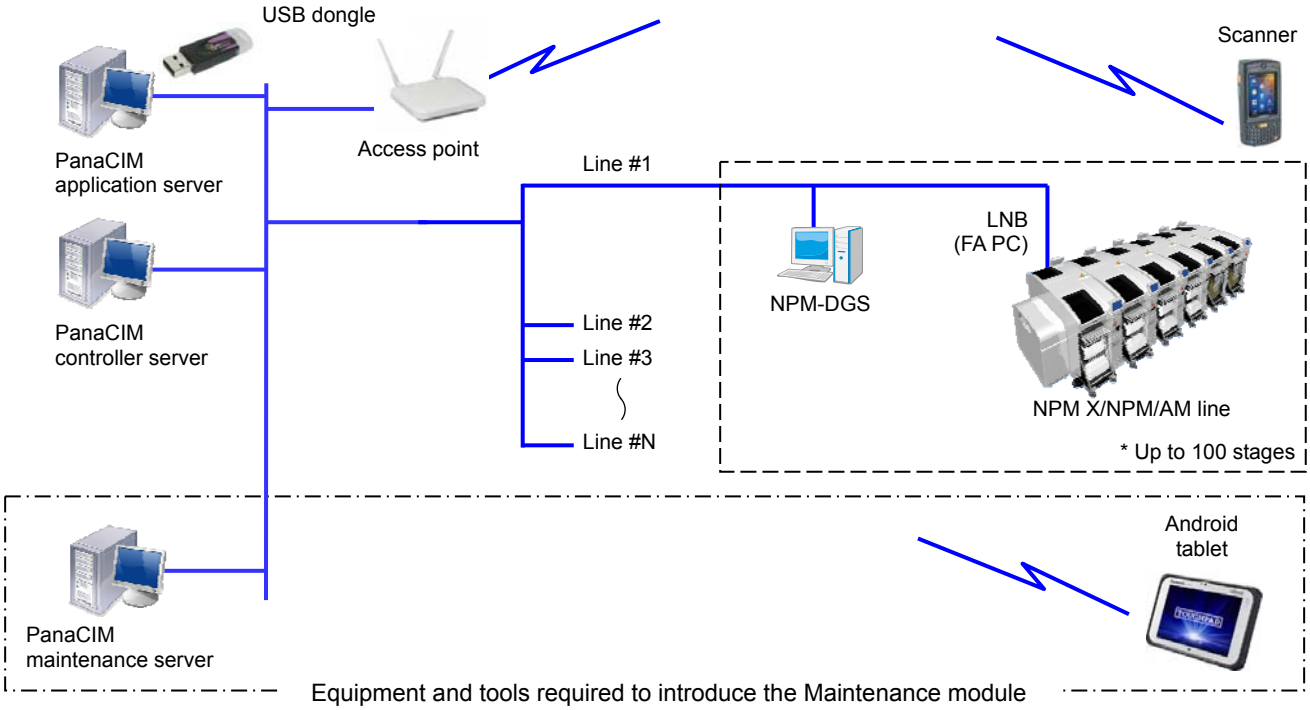


Large-scale (multiple lines) configurations of PanaCIM

One PanaCIM controller server can control up to 100 stages.

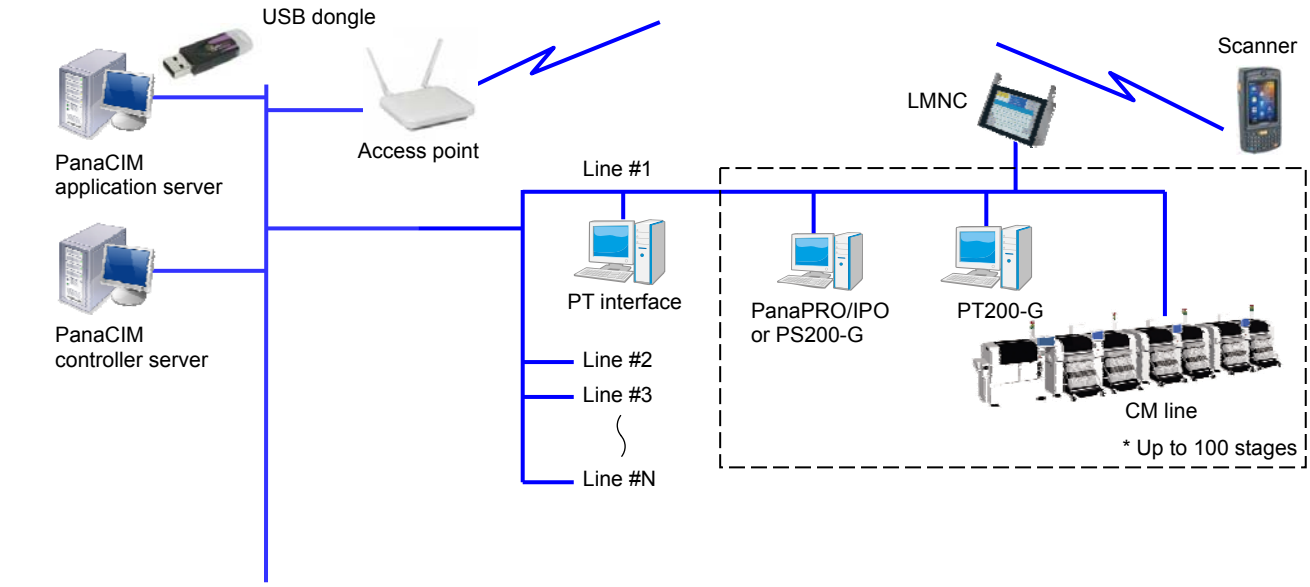
* Please install a Production Monitoring Client if the Production Monitoring Module is used and the number of stages is more than 7 per line and the number of lines is more than 4.
  The necessary number of pieces of equipment depends on the machine configuration. So, for details, please contact us.

For NPM X/NPM/AM line

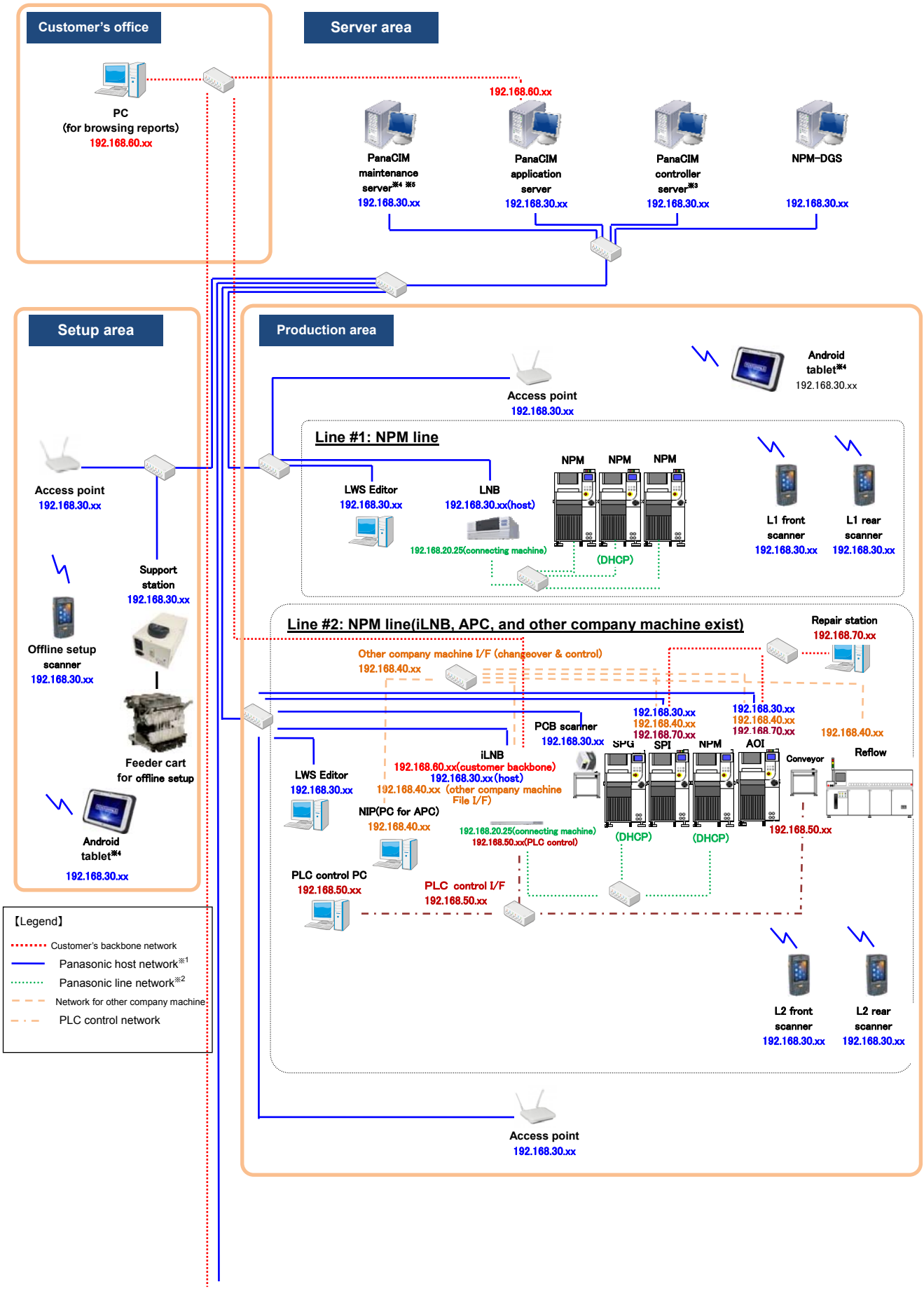


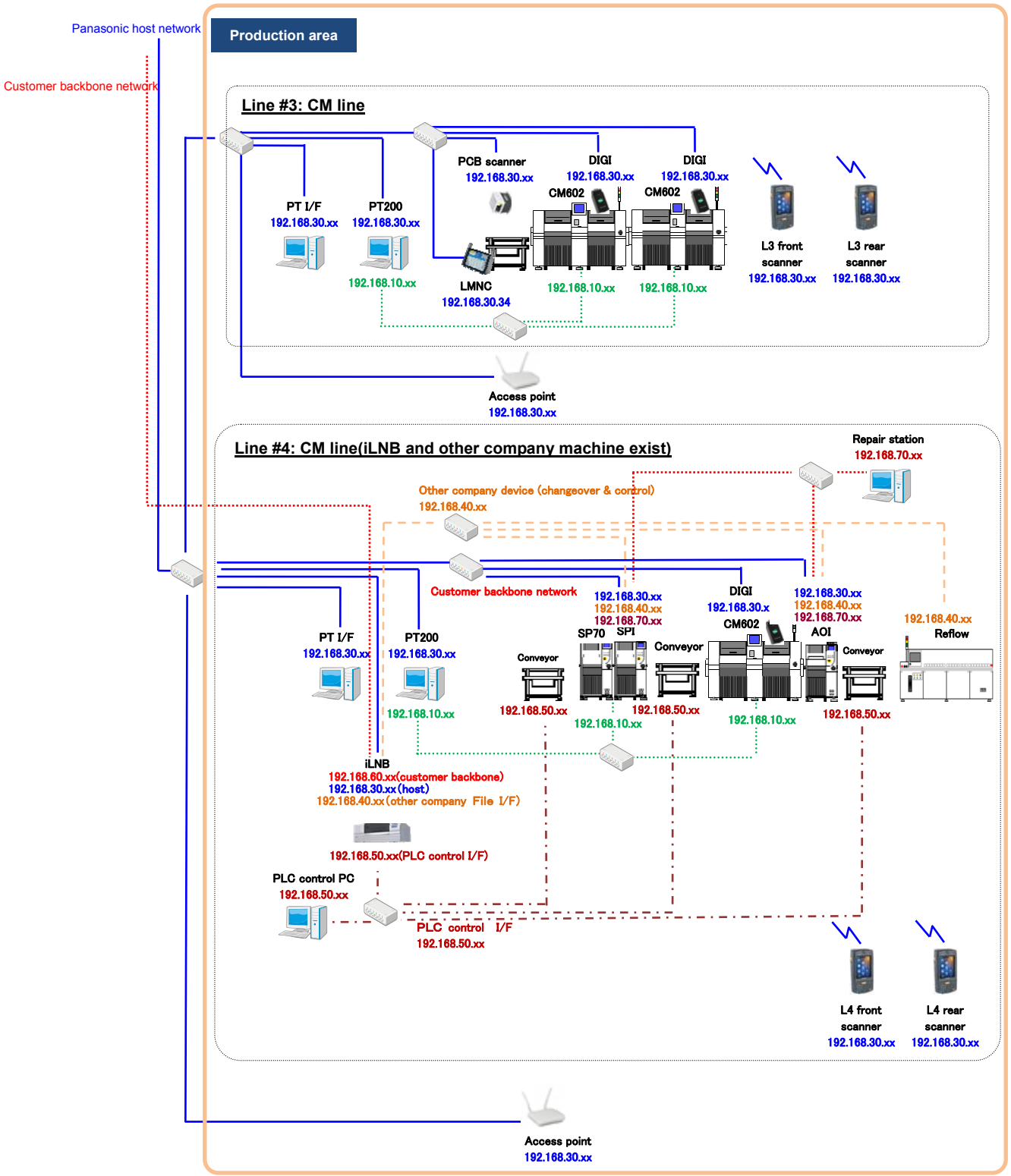
* Multiple NPM-DGS can be connected to one PanaCIM

For CM line



PanaCIM network configuration example




Panasonic host network
Customer backbone network
Production area
Line #3: CM line
PT I/F
192.168.30.xx
PT200
192.168.30.xx
192.168.10.xx
PCB scanner
192.168.30.xx
LMNC
192.168.30.34
DIGI
192.168.30.xx
CM602
192.168.10.xx
DIGI
192.168.30.xx
CM602
192.168.10.xx
L3 front scanner
192.168.30.xx
L3 rear scanner
192.168.30.xx
Access point
192.168.30.xx
Line #4: CM line(iLNB and other company machine exist)
Repair station
192.168.70.xx
Other company device (changeover & control)
192.168.40.xx
Customer backbone network
PT I/F
192.168.30.xx
PT200
192.168.30.xx
192.168.10.xx
Conveyor
192.168.50.xx
SP70
SPI
192.168.30.xx
192.168.40.xx
192.168.70.xx
Conveyor
192.168.50.xx
192.168.10.xx
DIGI
192.168.30.x
CM602
192.168.10.xx
192.168.30.xx
192.168.40.xx
192.168.70.xx
AOI
Conveyor
192.168.50.xx
192.168.40.xx
Reflow
iLNB
192.168.60.xx(customer backbone)
192.168.30.xx(host)
192.168.40.xx(other company File I/F)
192.168.50.xx(PLC control I/F)
PLC control PC
192.168.50.xx
PLC control I/F
192.168.50.xx
L4 front scanner
192.168.30.xx
L4 rear scanner
192.168.30.xx
Access point
192.168.30.xx

*1 IP addresses, 192.168.10.xx, 192.168.20.xx and 192.168.30.xx, 192.168.40.xx, 192.168.50.xx, are used in the Panasonic local network.
*2 The network connected to each line should be configured in a star. Note that if a network, configured in a ring or bus, is physically interrupted, other lines involved in the network will also be interrupted.
*3 In this example, just one PanaCIM controller server is enough to control the system because the number of stages is less than 100 (NPM: 1 stage x 6 + CM: 2 stages x 2 = 10 stages).
*4 Required only when introducing the Maintenance module
*5 Refer to 5.1 for the models for which the Maintenance module can be used. In this example, applicable model is NPM only.
*6 Prepare SW-HUB 1000BASE-TX as a hub.

System equipment configuration (Express package)

Constituent elements	OS / Database	Role
	Restrictions	
PanaCIM application server	Windows® 7 Professional (32-bit/ 64-bit version) Windows® 8.1 Professional (64-bit version) Microsoft® SQL Server® 2008 Express R2 (32/64-bit version) Microsoft® SQL Server® 2012 Express (32/64-bit version)	・Application host ・Provision of Web portal (Web server) ・License management ・Data control (Database)
	・Internet Explorer® 11 is required. *For IE8, turn ON “Compatibility View”.	
PanaCIM controller server	Cent OS 6.2 (32-bit version)	・Server that controls line configuration, various settings and the data, events or the like of all PanaCIM applications
	・Up to 5 stages are controllable	
PT interface	Windows® 7 Professional (32-bit/ 64-bit version) Windows® 8.1 Professional (64-bit version)	・Carries out data communication with PT200
Web client (Report reference PC)	Windows® 7 Professional (32-bit/ 64-bit version) Windows® 8.1 Professional (64-bit version)	・Accesses the PanaCIM application server and produces graph / report displays
	・A new purchase is not required. It is available with your existing PC that allows access to the PanaCIM application server. (The response speed and others depend on your network environment) ・You do not have to use the PanaCIM application server to access reports. ・Internet Explorer® 11 is required. *For IE8, turn ON “Compatibility View”.	
Access point*1	—	・Wireless communication between the PanaCIM controller server and the scanner
	・Required for Material Verification	
Wireless scanner*1	—	・Reads barcodes while in the Material Verification process
	・Required for Material Verification	

*1 Equipment prepared by you, which is required according to the module (function) you use.
The necessary number of pieces of equipment depends on the machine configuration. So, for details, please contact us.

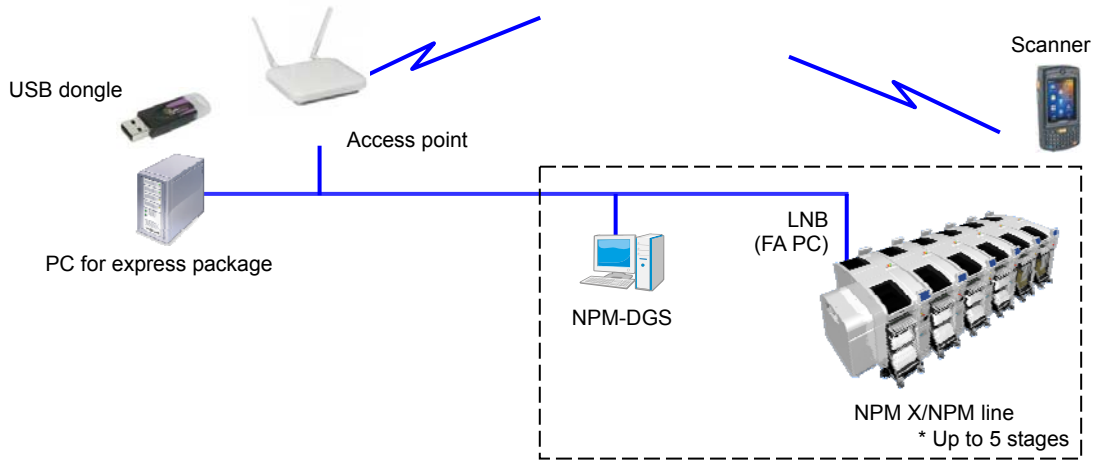
1 line configurations of PanaCIM (Express package)

PanaCIM express package is installable under the following conditions.

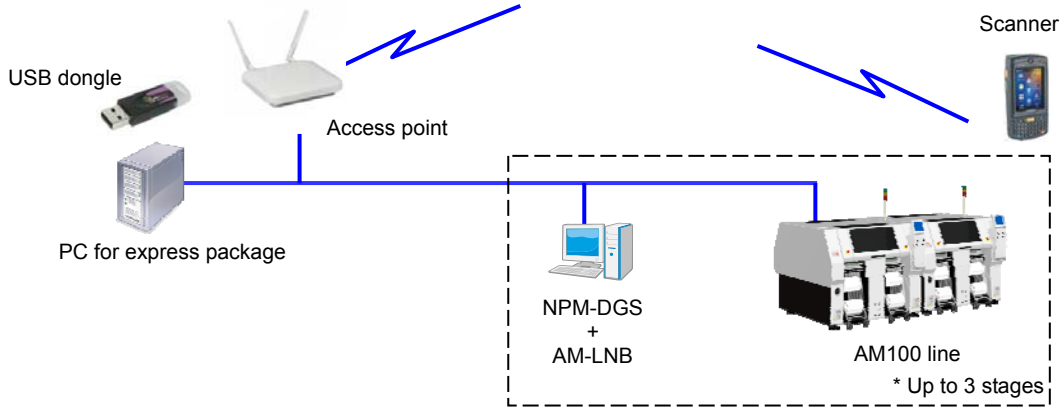
1) Up to 5 stages per line
2) Use only material verification module and production analysis module
3) Data retention period of production analysis: 30 days

* For PanaCIM express package, a PC with the following configuration is required per line.
DELL PowerEdge T130 Ⅱ (Memory: 8 GB, HDD: 500 GB)

For NPM X/NPM line

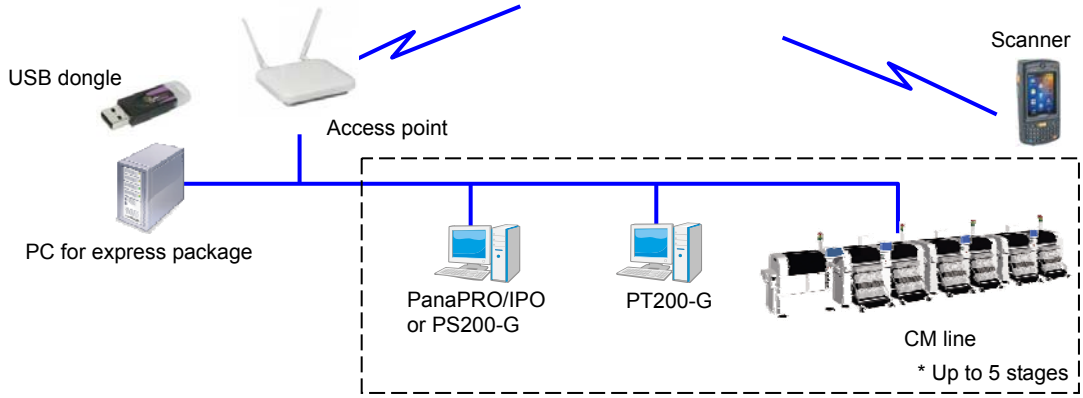


For AM100 line (Material verification only, Up to 3 stages)

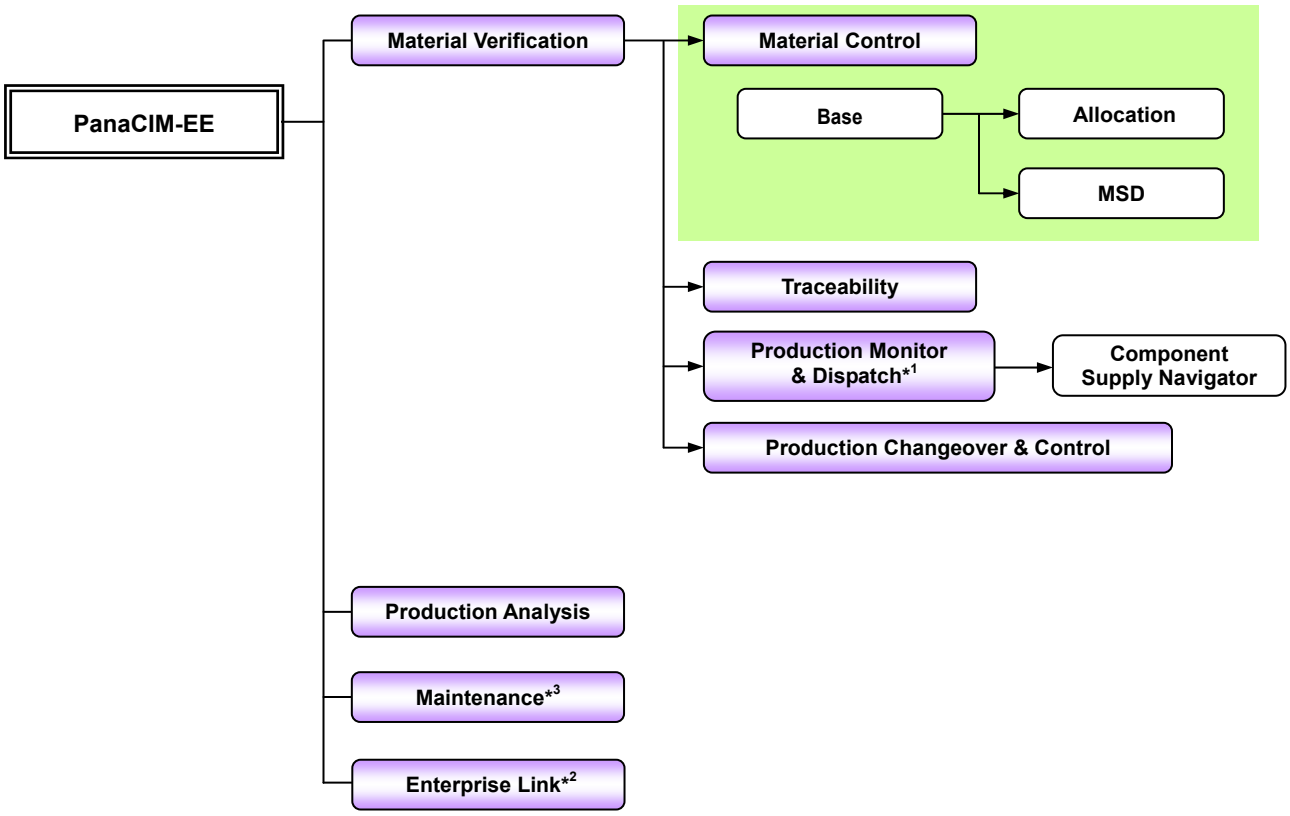


* If only material verification module, possible to use AM-LNB.
AM-LNB can be shared with NPM-DGS PC (Applicable OS: Windows® 7 Professional only)

For CM line

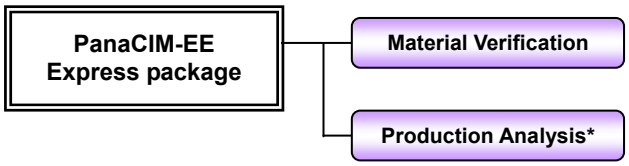


Module configuration



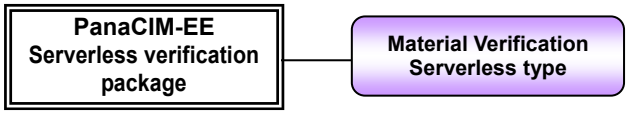
*1 You need a Production Analysis module license to display the operating information of the machine.
*2 Because data as an output source is required, you need any of Material Verification, Traceability and Production Analysis module licenses.
*3 License of Production Monitor & Dispatch may be required if need arises.

Module configuration (Express package)



* Data retention period of production analysis: 30 days

■ Module configuration (Serverless verification package)



3.3 Hardware / Software Specifications

Server, PC

Item		Specifications			
		PanaCIM application server	PanaCIM controller server	PT interface	PanaCIM Production Monitor & Dispatch client
Main computer unit		IBM PC/AT-compatible (A desktop PC is strongly recommended.)			
	CPU	Intel® Core™2 Duo E6700 equivalent or faster (For V8.6.0.1 or later, Intel® Xeon E3-1220 3.1 GHz equivalent or faster is strongly recommended.)		Intel® Core™ i3/i5/i7 (8 MB Cache, 3.9 GHz) equivalent or faster	
	Mother board	Fully IBM compatible			
	Serial I/O	Fully IBM compatible			
	Graphic board	SXGA or higher (Display: 65,536 colors or more) Desktop area: 1,024 × 768 dot or more			
	Memory	16 GB or more	4 GB or more	4 GB or more	
	HDD	500 GB or more	500 GB or more	250 GB or more	80 GB or more
		NTFS file system	Ext3 file system	NTFS file system	
	Optical drive	DVD drive			
Keyboard		106 keyboard			
Mouse		Supported by your OS as standard			
Monitor		SXGA or higher ready			
Network card		1000 BASE-T × 2			

PanaCIM server and PC operating environments

Item		Language	Application			
			PanaCIM application server	PanaCIM controller server	PT interface	PanaCIM Production Monitor & Dispatch client
OS						
	Windows Server® 2008 R2 (64-bit version)	English Japanese	○	—	—	—
	Windows Server® 2012 R2 (64-bit version)		○	—	—	—
	Windows Server® 2016 (64-bit version)		○	—	—	—
	Red Hat Enterprise Linux 5.2/5.4/5.7/6.5/6.7 Server (32-bit version)		—	○	—	—
	Windows® 7 Professional (32-/64-bit version) Windows® 8.1 Professional (64-bit version)		—	—	○	○
Other software						
	Microsoft® SQL Server® 2008 R2 Standard/Enterprise Edition (64-bit version)	English Japanese	○	—	—	—
	Microsoft® SQL Server® 2012 R2 Standard/Enterprise Edition (64-bit version)		○	—	—	—
	Microsoft® SQL Server® 2016 Standard Edition (64-bit version)		○	—	—	—
	Internet Explorer® 11 is required. * For IE8, turn ON “Compatibility View”.		○	—	○	○
	Microsoft® .NET 3.5		○	—	—	—
	IIS 6.0		○	—	—	—
	Microsoft's Message Queuing Service		○	—	—	—

○: Applicable software, —: No applicable software

■ **Server and PC: To run the Maintenance module, hardware satisfying the following requirements must be prepared besides the PanaCIM application server and controller server.**

Item		Specifications		
		PanaCIM maintenance server	PanaCIM maintenance client	Mobile device
Main computer unit		IBM PC/AT-compatible (A desktop PC is strongly recommended.)		Android tablet
	CPU	Intel® Xeon® E3 (4 core) equivalent or faster	Intel® Core™ i3/i5/i7 (8 MB Cache, 3.9GHz) equivalent or faster	Intel® Atom™ x3 or equivalent Qualcomm® Snapdragon™ S4 equivalent or faster
	Mother board	Fully IBM compatible		—
	Serial I/O	Fully IBM compatible		—
	Graphic board	SXGA or higher (Display: 65,536 colors or more) Desktop area: 1,280 × 1,024 dot or more		7-inch color LCD, WXGA or better, Display:65,536 colors or more
	Memory	16 GB or more	4 GB or more	2 GB or more
	HDD	1.0 TB or more	80 GB or more	32 GB or more
		NTFS file system		—
	Optical drive	DVD drive		—
Keyboard		106 keyboard		—
Mouse		Supported by your OS as standard		Capacitive multi-touch
Monitor		SXGA or higher ready		—
Network card		—		Back: 8 megapixels or more
Keyboard		1000 BASE-T × 2		IEEE802.11a (W52/W53/W56)/b/g/n-compliant

■ **PanaCIM servers and PC operating environments: To run the Maintenance module, software satisfying the following requirements must be prepared besides those required for the PanaCIM application server and controller server.**

Item		Language	Application		
			PanaCIM maintenance server	PanaCIM maintenance client	Android tablet
OS					
	Windows Server® 2012 R2 (64-bit version)	English Japanese	○	—	—
	Windows Server® 2016 (64-bit version)		○	—	—
	Windows® 7 Professional (64-bit version) Windows® 10 Pro (64-bit version)		—	○	—
	AndroidTM 6.0 or later		—	—	○
Other software					
	Microsoft® SQL Server® 2012 R2 Standard/Enterprise Edition (64-bit version)	English Japanese	○	—	—
	Microsoft® SQL Server® 2016 Standard Edition (64-bit version)		○	—	—
	Internet Explorer11 * Set to Compatibility View for IE8 before use.		○	○	—
	Google Chrome™		—	○	○
	Microsoft® .NET 3.5		○	—	—
	IIS 6.0		○	—	—
	Microsoft's Message Queuing Service		○	—	—

○: Applicable software, —: No applicable software

Server (Single server configuration)

Item		Specifications	
		No maintenance module (PanaCIM application server + PanaCIM control server)	Maintenance module included
Main computer unit		IBM PC/AT-compatible (A desktop PC is strongly recommended.)	
	CPU	Intel® Xeon® E5 (6 core) equivalent or faster	Intel® Xeon® E5 (8 core) equivalent or faster
	Nother board	Fully IBM compatible	
	Serial I/O	Fully IBM compatible	
	Graphic board	SXGA or higher (Display: 65,536 colors or more) Desktop area: 1,024 × 768 dot or more	
	Memory	20 GB or more	36 GB or more
	HDD	1.0 TB or more	2 TB or more
		NTFS file system	NTFS file system
	Optical drive	DVD drive	
Keyboard		106 keyboard	
Mouse		Supported by your OS as standard	
Monitor		SXGA or higher ready	
Network card		1000 BASE-T × 2	
USB port		USB port × 1	

■ Server (Single server configuration) operating environments

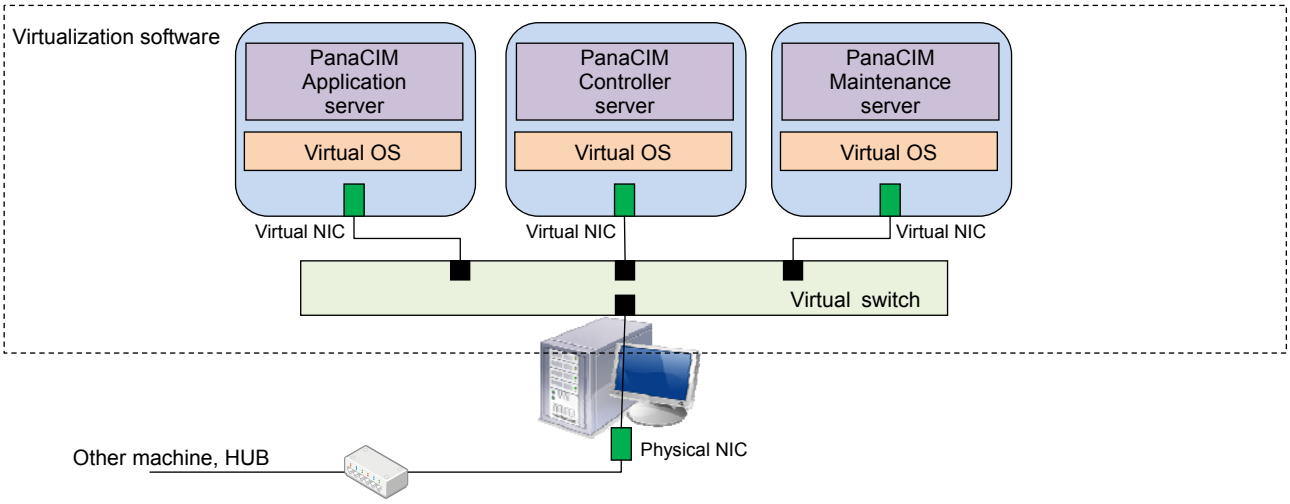
Item	Language	Application	
		No maintenance module (PanaCIM application server + PanaCIM control server)	Maintenance module included
Virtualization software	English Japanese		
VMware ESXi 6.0		○	

○: Applicable software, －: No applicable software

* In addition to the above software, software required to run the PanaCIM application server, PanaCIM controller server and PanaCIM maintenance server also must be installed.

* With a minimum purchased license (16 core licenses) for Windows Server 2016, single server configuration (When using a plurality of pieces of virtual machine on virtualization software) can use two virtual machines on one physical server.
Therefore, a minimum purchased license for Windows Server 2016 can run both PanaCIM Application Server and PanaCIM Maintenance Server.

■ Network configuration in single server configuration



* When virtualization software constructs a plurality of pieces of application in single server configuration, each virtualization OS can have virtual NIC (network card). This can consolidate into one physical NIC.
Providing customer backbone network requires a plurality of pieces of physical NIC.

PC for Express package

For PanaCIM express type, a PC with the following configuration is required per line.
DELL　PowerEdge T130Ⅱ(Memory: 8 GB, HDD: 500 GB)

PC operating environments (For Express package)

Item		Language	Application		
			PanaCIM application server	PanaCIM controller server	PT interface
OS					
	Windows® 7 Professional (32-/64-bit version) Windows® 8.1 Professional (64-bit version)	English Japanese	○	―	○
	Cent OS 6.2 (32-bit version)		―	○	―
Other software					
	Microsoft® SQL Server® 2008 Express R2 (32-/64-bit version) Microsoft® SQL Server® 2012 Express R2 (32-/64-bit version)	English Japanese	○	―	―
	Internet Explorer11 * Set to Compatibility View for IE8 before use.		○	―	○
	Microsoft® .NET 3.5		○	―	―
	IIS 6.0		○	―	―
	Microsoft's Message Queuing Service		○	―	―
Virtualization software		English Japanese			
	VMWare ESXi V6.0		○		

○: Applicable software, ―: No applicable software

Equipment necessary for using Material Verification, Material Control and Production Monitor & Dispatch modules (Equipment to be provided by customer)

Item	Description		
	Equipment / Model No.*2 *4		Class
Wireless scanner (Motorola, Inc.)	MC55*1 	QWERTY keyboard type (2-dimensions) MC55 Model No.: MC55E0-PL0S3QQA9WR※3 Model No. (South Korea & China): MC55E0-PL0S3QQA9WR	Indispensable
	Cradle unit 	Model No. (MC55): MC5X-RCHG1-01-SET※3	Indispensable
		USB/serial single slot cradle	
		AC adapter	
		Power cable	
		USB communication cable	
	Spare battery	・2,400 mAh Standard battery (For MC55) Model No.: BTRY-MC55EAB02	Recommended
Access point (ZEBRA)	AP7522 	Model No.: AP-7522-67040-APME-AC—SET*3	Indispensable
		Wireless LAN access point	
		Dual band antenna	
		AC adapter	
		Power cable	
		Operating Instructions (CD)	
Switching hub	1000BASE-T-ready		
LAN cable	Category 5 or higher		
Digi One	Model No.：N510002864AA Type：Digi One IA 70001862		Required for CM-based machine

*1 The wireless scanner comes with a standard battery, battery cover with hand strap, and stylus pen.
*2 Model numbers are subject to change by the manufacturer, so you are encouraged to check model numbers at the time of purchase.
*3 Please prepare power cables and related equipment that are suitable for use in your local area.
*4 Applicable also when “Serverless package” is used.

Equipment necessary for using Traceability and Production Changeover & Control modules (to be provided by customer)

Item	Description		
	Equipment	Model No.*2	Necessity
Installed barcode reader (IDEC DATALOGIC)	1D / 2D fixed barcode reader (for PCB ID read only)*1	Matrix-2000	Choice
Installed barcode reader (Mini-Hawk)	1D / 2D fixed barcode reader (For PCB ID read only)*1	FIS-6300-5200G	
Installed barcode reader (Keyence)	1D / 2D fixed barcode reader (For PCB ID read only)*1	SR600*3 SR750	
LMNC*4 (Elcon) 	CM line control node controller	PanaCIM-EE LMNC	Required for CM-/ SP-based machine

*1 Production Changeover & Control for the NPM X/NPM/AM series are controlled by the machine (external scanner read type, head read type, planning report read type). (For details, refer to “Specification” booklet of each model.)
*2 Model numbers are subject to change by the manufacturer, so you are encouraged to check model numbers at the time of purchase.
*3 Ethernet communication unit is needed separately for the NPM series and AM100.
*4 Please contact our sales representatives for the purchase of it.

Equipment necessary for using Maintenance module (to be provided by customer)

<table>
<tr><th rowspan="2">Item</th><th colspan="3">Description</th></tr>
<tr><th colspan="2">Equipment / Model No.※1</th><th>Class</th></tr>
<tr><td>Android tablet</td><td>FZ-B2D*1
  </td><td colspan="2">Prepare a tablet that satisfies the following requirements.
CPU: Equivalent to or faster than Intel® Atom™ x3 or Qualcomm®SnapdragonTM S4
RAM: 2GB or more
HDD: 32GB or more
Screen: 7-inch color LCD, WXGA or better, 65,356 colors or more
Capacitive multi-touch
Camera: Back: 8 megapixels or more
Network: IEEE802.11a(W52/W53/W56)/b/g/n-compliant
OS: Android 6.0 or later</td><td>Indispensable</td></tr>
<tr><td>Access point
(ZEBRA)</td><td>AP7522
  </td><td colspan="2">Model: AP-7522-67040-APME-AC—SET*3
Wireless LAN access point
Dual band antenna (x 3)
AC adapter
Power cable
Operating Instructions (CD)</td><td>Indispensable</td></tr>
<tr><td>Switching hub</td><td colspan="2">1000BASE-T-ready</td><td></td></tr>
<tr><td>LAN cable</td><td colspan="2">Category 5 or higher</td><td></td></tr>
</table>

*1 Model number differs from country to country. Prepare a model appropriate for use in your country.

3.4 Specification Necessary for Machine

Specification necessary for using Material Verification, Material Control and Production Monitor & Dispatch modules

Item		Description		
		Option name	Remarks	Necessity
NPM X/NPM/AM series		PanaCIM-EE ready	License	Indispensable
	Support station	Support station box	With wired scanner*2	Required for off-line setup
CM series*1 machines		PanaPRO/CVT supporting unit: For MC70/ MC75	Cradle holder Power supply Communication unit	Indispensable
	Change Cart Preparation Unit	Cradle holder Communication unit	Cradle holder Communication unit	Required for off-line setup
Screen printers*5		PanaCIM-EE ready	License	Indispensable
Common		Intelligent Tape Feeder*3	With the joint detection sensor	Indispensable
		Splicing tape*4	Black tape	Indispensable

*1 CM602, CM101, CM232, CM212, and CM402, CM401/ DT401 series
*2 The wireless scanner (MC55) is used for material verification.
*3 Not compatible with bulk, stick and 32 mm adhesive feeders.
*4 Performance cannot be assured with any splicing tape other than Panasonic products.
*5 Screen printers supported by LNB (SP70, SPG, SPD, SPV, etc.)

Specification required for using Traceability, Production Analysis and Enterprise Link modules

Item	Description		
	Option name	Remarks	Necessity
NPM X/NPM/AM series, Screen printers*1	PanaCIM-EE ready	License	Indispensable

*1 Screen printers supported by LNB (SP70, SPG, SPD, SPV, etc.)

Specification required for using maintenance module

Item	Description		
	Option name	Remarks	Necessity
NPM X/ NPM / AM series machines	PanaCIM-EE ready	License	Indispensable

**** Remarks ****

- Refer to “5.1 Target machines” for each module.

4. Module Function

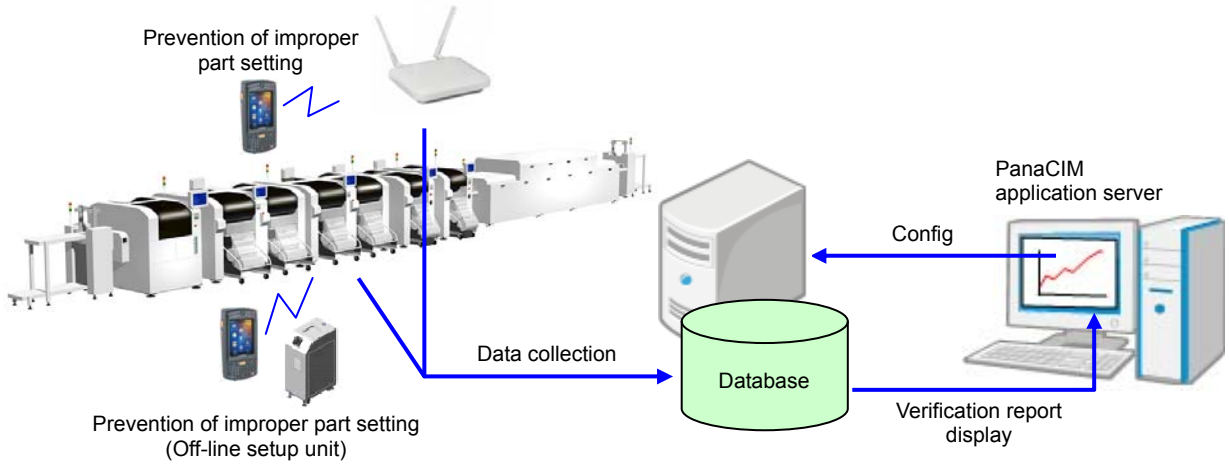
4.1 Material Verification Module

A support function, performed in the case of setup for production/ product changeover/ splicing/ parts shortages, which compares (verifies) material or parts information to (and) production programs before placement of materials or parts and thereby prevents the wrong materials or parts from being placed in the machine.

Main functions

(1) Compares and verifies material/ parts information read by the barcode reader and that of production programs.
(2) Displays a warning message when a material or part not used in production is scanned , or when a material or part not used in production is found to have been set, and also controls the start of machine production*[1].
(3) Displays a report about a history of material/ parts verification (a scan history of material/ parts verification).

*1 As to controlling the start of machine production, target models are subject to some constraints (e.g., the printer is excluded). For details, contact us.



Function of Material Verification Module

Main function	Detailed function		Description
	Large category	Small category	
Function to prevent improper use of materials	Part replacement procedure		Allows verifying parts when replacing them
	Production start control procedure		Uses the production start control procedure to reduce production when the wrong parts are found to be set
	Alternate parts		Alternate parts can be used in place of original parts already defined Both original and alternate parts can be recorded for use in trace
	Invalid parts		Invalid parts can be specified. Also, invalid parts, if scanned, are prevented from being installed
	Tray parts		Parts that can be verified when replaced
	New / old reel barcode comparison		Only when the part name of an old part reel (currently in use) matches that of a new part reel (to be fed for splicing) before splicing, splicing verification will be authorized. Additionally, a single comparison item, except for parts name, can be added (Any of Lot, Vendor and a User-defined item is the same or different)
	Regular verification check		Allows verification with given timing or on a periodic basis
	Machine interlock		Disallow the start of production with verification not completed
	Off-line setup verification		Reduces the amount of time required for changeover by performing advance parts verification before the start of production
	Feeder free arrangement		Allows verification of feeders that, in sync with the feeder free arrangement function of the machine, were placed in positions different from those assumed by a production program
	Materials for screen printer		Allows verification of solder, masks, squeegees, etc.*[1]
	Common	Scanner	Can read 1D or 2D barcodes with a scanner
			Allows verification of multiple lines / off-line setup with a single scanner
		Barcode format	The following barcode formats are supported; 1D code (Barcode): Code 39, Code 128, Code 93, Code 2of5, Interleaved, UPC, EAN, CODEBAR, MSI 2D code: Data Matrix, QR Code, PDF417
		Operator ID	Can read operator IDs and record an operating history during verification

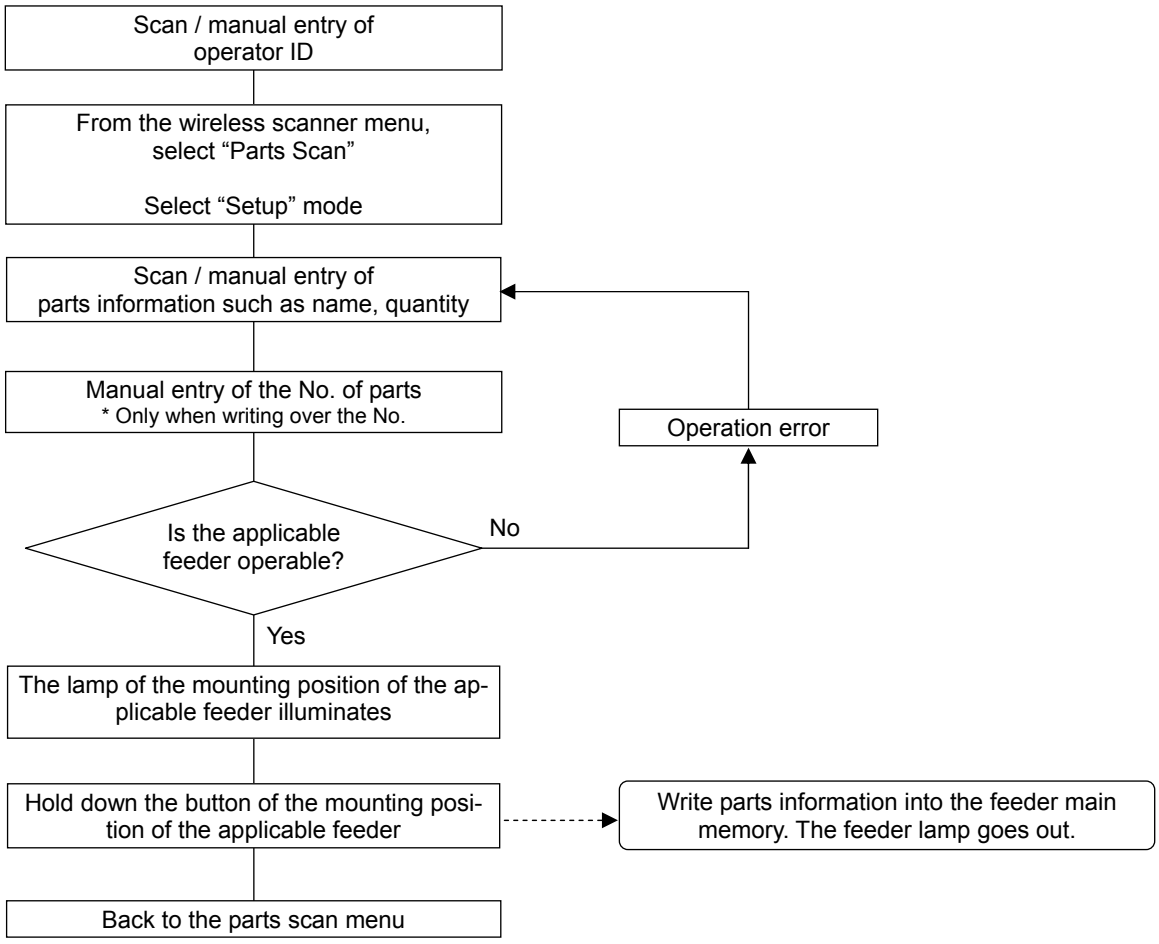
Material verification module function (Continued)

Main function	Detailed function		Description
	Large category	Small category	
Data collection function	Data type		Part No., Lot, Vendor, Quantity, User-defined item
	Number of machines which can collect data		100 stages (per PanaCIM controller server) * For express package: Up to 5 stages (Use AM-LNB: Up to 3 stages)
	Number of lines which can collect data		The equivalent of the number of stages above (per PanaCIM controller server) * For express package: Per line
	Collection interval		Every time a part is scanned
Data storage function	Backup function	Automatic backup	Allows data to be backed up automatically at setting intervals
		External storage	Backup data can also be stored in external media (other media)
	Restoration function	Restore	Data can be restored based on backup data
	Report display	Type	Reports can be displayed
		Save	Parameters (attribute value, range, etc.) necessary for preparing reports can be saved as report configuration files
		Quick-run	A report can be displayed by calling up the report configuration file saved
		Display time	After report settings are configured, reports can be displayed in a short time by carrying out report generation or quick run
Setting function	Report settings		Can select a report you want to display, and set parameters including time intervals and graph types
	Barcode settings		Can change the settings for parts information barcodes

*1 Currently, SPV, SPD, SPG, and SP70 (LNB connection) are subject to the following constraints

- The line configuration shall be the same as NPM X, NPM, and AM100 series (Line configuration with a single machine is not supported)
  - Verification of alternative parts is not supported
  - Only one SP70 is supported per line

Work flowchart of material verification (Flowchart for reel parts at the time of setup)

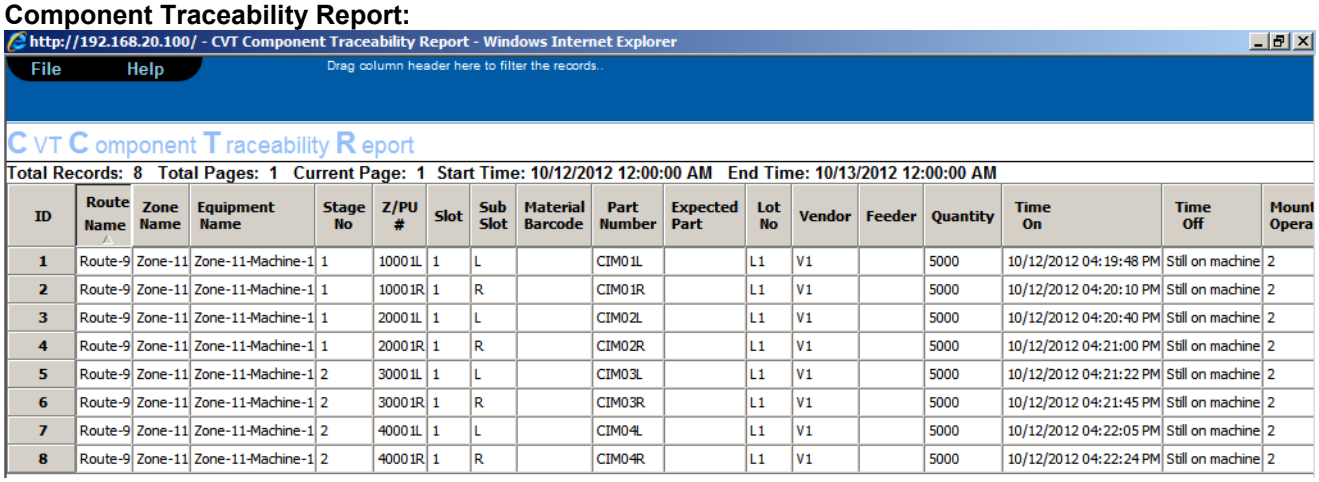


Material verification module Report function

Material and parts verification report		
	Component Trace	In the case of PCBs with certain materials/ parts (narrow down under specific conditions such as parts number, vendor code, Lot, Reel, Barcode, etc.) mounted, or paste/ mask, the list of printed PCBs is displayed
	Current Setup Report	Displays setup information as of now (part reel information).
	History Report	Displays a list of histories that a user has verified (parts shortages, feeder jammed, setup, etc.) for a specified period of time
	Transaction Trace Report	Displays the history of parts verification operations (parts shortages, feeder jammed, setup, etc.) / related events (splicing detection)
Production setup		
	Product setup information	Allows displaying the setup conditions (model ID allocation, equipment allocation) of production programs
	Mix information	Allows displaying the relation of mix and production programs
	Changeover	Displays a list of the feeders required for production of the selected model and their positions on a per machine basis.
Advance setup report		
	Advance setup (Details)	Displays a detailed list of advance setups of the selected machine
	Advance setup (Overview)	Displays an outline list of advance setups of the selected machine
	Advance setup cart Installation history	Displays a history of verifications operated and advance setups done with carts for a specified period of time
	Advance setup cart Current setup	Displays current setup information of the advance setup cart
	Advance setup cart Feeder history	Displays a history of feeders inserted and removed with carts for a specified period of time
	Advance setup cart Installation status	Displays a history of verifications operated and advance setups done with carts for a specified period of time
	Advance setup cart Check	Displays the position (table) of the advance setup cart being installed to the machine
	Advance setup cart Data confirmation	Displays the advance setup status (the No. of verifications needed and the No. of verifications)

Example of a material verification report

Component Traceability Report:



http://192.168.20.100/ - CVT Component Traceability Report - Windows Internet Explorer

File Help

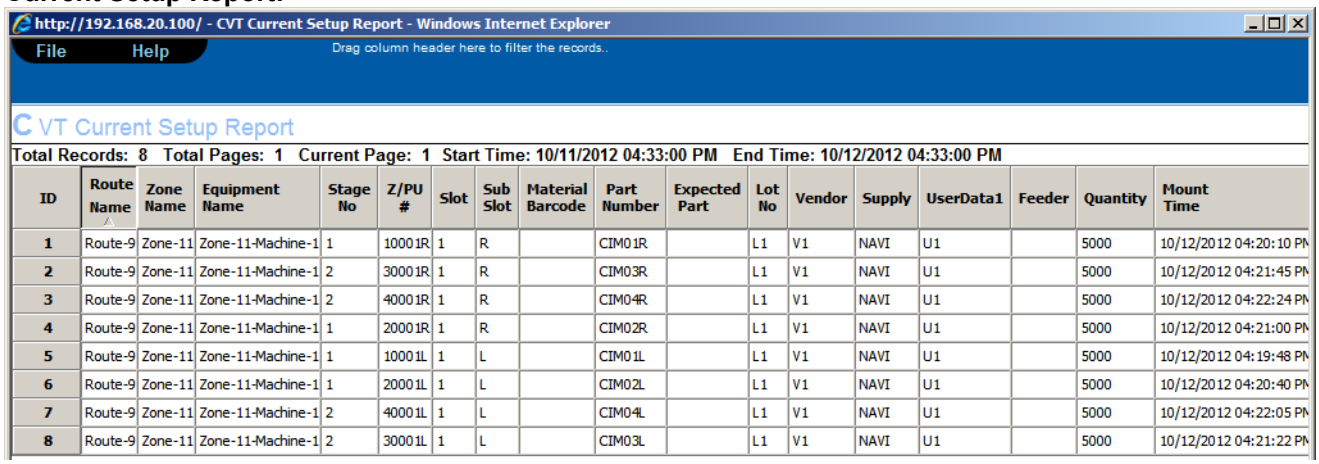
Drag column header here to filter the records..

CVT Component Traceability Report

Total Records: 8 Total Pages: 1 Current Page: 1 Start Time: 10/12/2012 12:00:00 AM End Time: 10/13/2012 12:00:00 AM

ID	Route Name	Zone Name	Equipment Name	Stage No	Z/PU #	Slot	Sub Slot	Material Barcode	Part Number	Expected Part	Lot No	Vendor	Feeder	Quantity	Time On	Time Off	Mount Opera
1	Route-9	Zone-11	Zone-11-Machine-1	1	10001L	1	L		CIM01L		L1	V1		5000	10/12/2012 04:19:48 PM	Still on machine	2
2	Route-9	Zone-11	Zone-11-Machine-1	1	10001R	1	R		CIM01R		L1	V1		5000	10/12/2012 04:20:10 PM	Still on machine	2
3	Route-9	Zone-11	Zone-11-Machine-1	1	20001L	1	L		CIM02L		L1	V1		5000	10/12/2012 04:20:40 PM	Still on machine	2
4	Route-9	Zone-11	Zone-11-Machine-1	1	20001R	1	R		CIM02R		L1	V1		5000	10/12/2012 04:21:00 PM	Still on machine	2
5	Route-9	Zone-11	Zone-11-Machine-1	2	30001L	1	L		CIM03L		L1	V1		5000	10/12/2012 04:21:22 PM	Still on machine	2
6	Route-9	Zone-11	Zone-11-Machine-1	2	30001R	1	R		CIM03R		L1	V1		5000	10/12/2012 04:21:45 PM	Still on machine	2
7	Route-9	Zone-11	Zone-11-Machine-1	2	40001L	1	L		CIM04L		L1	V1		5000	10/12/2012 04:22:05 PM	Still on machine	2
8	Route-9	Zone-11	Zone-11-Machine-1	2	40001R	1	R		CIM04R		L1	V1		5000	10/12/2012 04:22:24 PM	Still on machine	2

Current Setup Report:



http://192.168.20.100/ - CVT Current Setup Report - Windows Internet Explorer

File Help

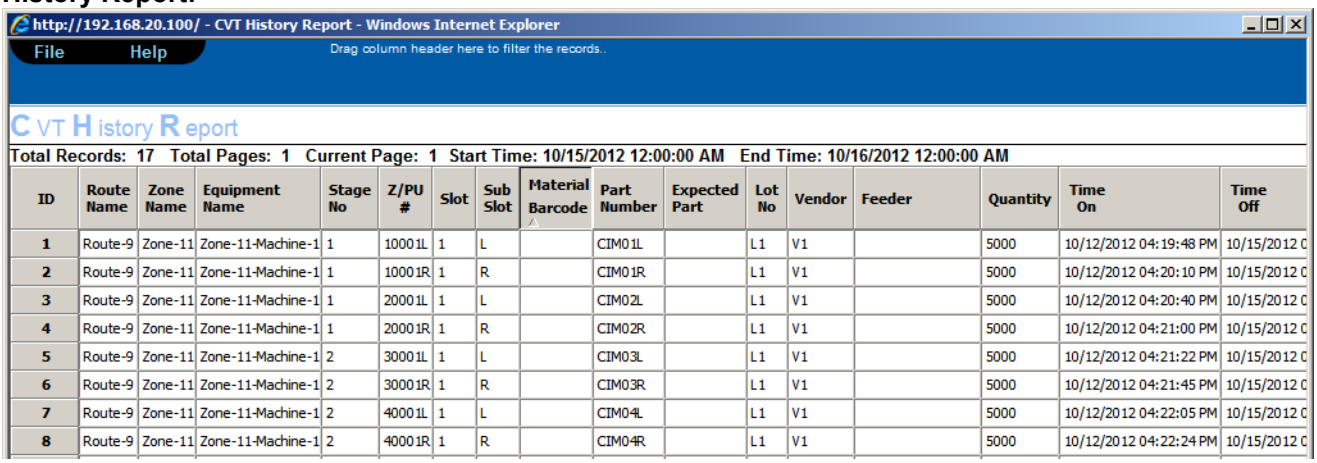
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CVT Current Setup Report

Total Records: 8 Total Pages: 1 Current Page: 1 Start Time: 10/11/2012 04:33:00 PM End Time: 10/12/2012 04:33:00 PM

ID	Route Name	Zone Name	Equipment Name	Stage No	Z/PU #	Slot	Sub Slot	Material Barcode	Part Number	Expected Part	Lot No	Vendor	Supply	UserData1	Feeder	Quantity	Mount Time
1	Route-9	Zone-11	Zone-11-Machine-1	1	10001R	1	R		CIM01R		L1	V1	NAVI	U1		5000	10/12/2012 04:20:10 PM
2	Route-9	Zone-11	Zone-11-Machine-1	2	30001R	1	R		CIM03R		L1	V1	NAVI	U1		5000	10/12/2012 04:21:45 PM
3	Route-9	Zone-11	Zone-11-Machine-1	2	40001R	1	R		CIM04R		L1	V1	NAVI	U1		5000	10/12/2012 04:22:24 PM
4	Route-9	Zone-11	Zone-11-Machine-1	1	20001R	1	R		CIM02R		L1	V1	NAVI	U1		5000	10/12/2012 04:21:00 PM
5	Route-9	Zone-11	Zone-11-Machine-1	1	10001L	1	L		CIM01L		L1	V1	NAVI	U1		5000	10/12/2012 04:19:48 PM
6	Route-9	Zone-11	Zone-11-Machine-1	1	20001L	1	L		CIM02L		L1	V1	NAVI	U1		5000	10/12/2012 04:20:40 PM
7	Route-9	Zone-11	Zone-11-Machine-1	2	40001L	1	L		CIM04L		L1	V1	NAVI	U1		5000	10/12/2012 04:22:05 PM
8	Route-9	Zone-11	Zone-11-Machine-1	2	30001L	1	L		CIM03L		L1	V1	NAVI	U1		5000	10/12/2012 04:21:22 PM

History Report:



http://192.168.20.100/ - CVT History Report - Windows Internet Explorer

File Help

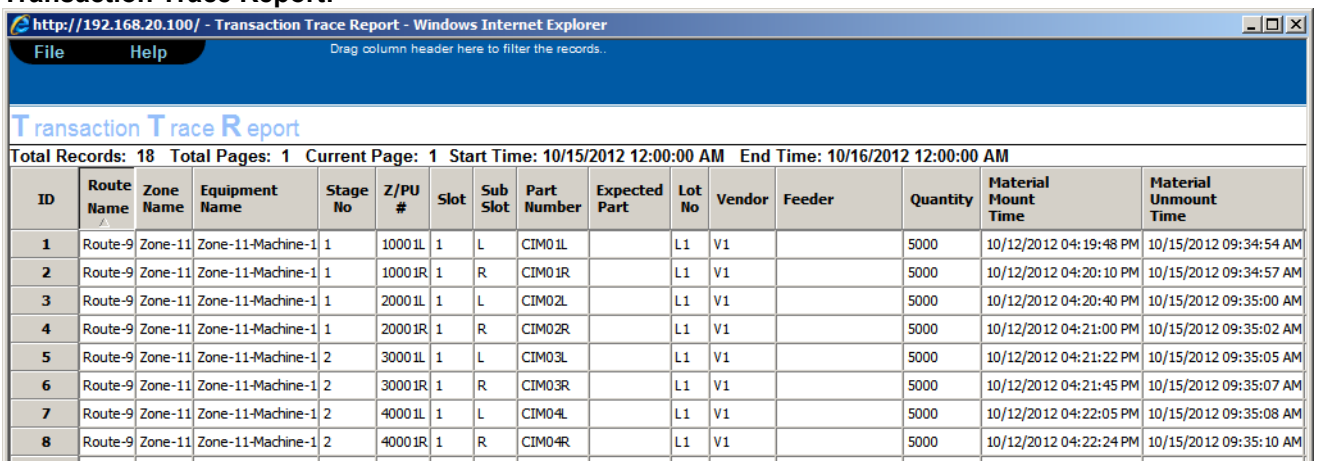
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CVT History Report

Total Records: 17 Total Pages: 1 Current Page: 1 Start Time: 10/15/2012 12:00:00 AM End Time: 10/16/2012 12:00:00 AM

ID	Route Name	Zone Name	Equipment Name	Stage No	Z/PU #	Slot	Sub Slot	Material Barcode	Part Number	Expected Part	Lot No	Vendor	Feeder	Quantity	Time On	Time Off
1	Route-9	Zone-11	Zone-11-Machine-1	1	10001L	1	L		CIM01L		L1	V1		5000	10/12/2012 04:19:48 PM	10/15/2012 0
2	Route-9	Zone-11	Zone-11-Machine-1	1	10001R	1	R		CIM01R		L1	V1		5000	10/12/2012 04:20:10 PM	10/15/2012 0
3	Route-9	Zone-11	Zone-11-Machine-1	1	20001L	1	L		CIM02L		L1	V1		5000	10/12/2012 04:20:40 PM	10/15/2012 0
4	Route-9	Zone-11	Zone-11-Machine-1	1	20001R	1	R		CIM02R		L1	V1		5000	10/12/2012 04:21:00 PM	10/15/2012 0
5	Route-9	Zone-11	Zone-11-Machine-1	2	30001L	1	L		CIM03L		L1	V1		5000	10/12/2012 04:21:22 PM	10/15/2012 0
6	Route-9	Zone-11	Zone-11-Machine-1	2	30001R	1	R		CIM03R		L1	V1		5000	10/12/2012 04:21:45 PM	10/15/2012 0
7	Route-9	Zone-11	Zone-11-Machine-1	2	40001L	1	L		CIM04L		L1	V1		5000	10/12/2012 04:22:05 PM	10/15/2012 0
8	Route-9	Zone-11	Zone-11-Machine-1	2	40001R	1	R		CIM04R		L1	V1		5000	10/12/2012 04:22:24 PM	10/15/2012 0

Transaction Trace Report:



http://192.168.20.100/ - Transaction Trace Report - Windows Internet Explorer

File Help

Drag column header here to filter the records..

Transaction Trace Report

Total Records: 18 Total Pages: 1 Current Page: 1 Start Time: 10/15/2012 12:00:00 AM End Time: 10/16/2012 12:00:00 AM

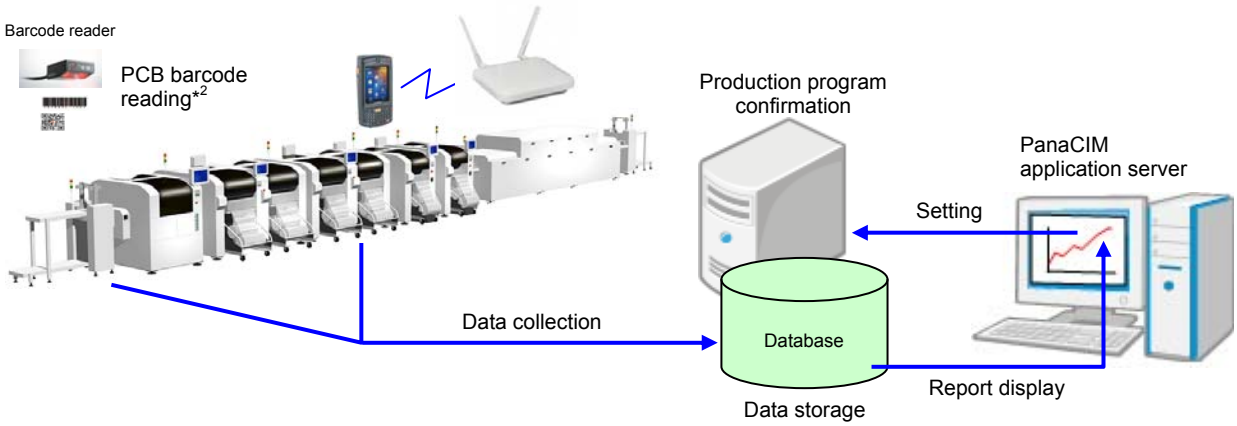
ID	Route Name	Zone Name	Equipment Name	Stage No	Z/PU #	Slot	Sub Slot	Part Number	Expected Part	Lot No	Vendor	Feeder	Quantity	Material Mount Time	Material Unmount Time
1	Route-9	Zone-11	Zone-11-Machine-1	1	10001L	1	L	CIM01L		L1	V1		5000	10/12/2012 04:19:48 PM	10/15/2012 09:34:54 AM
2	Route-9	Zone-11	Zone-11-Machine-1	1	10001R	1	R	CIM01R		L1	V1		5000	10/12/2012 04:20:10 PM	10/15/2012 09:34:57 AM
3	Route-9	Zone-11	Zone-11-Machine-1	1	20001L	1	L	CIM02L		L1	V1		5000	10/12/2012 04:20:40 PM	10/15/2012 09:35:00 AM
4	Route-9	Zone-11	Zone-11-Machine-1	1	20001R	1	R	CIM02R		L1	V1		5000	10/12/2012 04:21:00 PM	10/15/2012 09:35:02 AM
5	Route-9	Zone-11	Zone-11-Machine-1	2	30001L	1	L	CIM03L		L1	V1		5000	10/12/2012 04:21:22 PM	10/15/2012 09:35:05 AM
6	Route-9	Zone-11	Zone-11-Machine-1	2	30001R	1	R	CIM03R		L1	V1		5000	10/12/2012 04:21:45 PM	10/15/2012 09:35:07 AM
7	Route-9	Zone-11	Zone-11-Machine-1	2	40001L	1	L	CIM04L		L1	V1		5000	10/12/2012 04:22:05 PM	10/15/2012 09:35:08 AM
8	Route-9	Zone-11	Zone-11-Machine-1	2	40001R	1	R	CIM04R		L1	V1		5000	10/12/2012 04:22:24 PM	10/15/2012 09:35:10 AM

4.2 Traceability Module

A support function, designed to help control quality and meet market requirements, which controls history information on fabricated PCBs through the combined use of information such as the production performance record of the machine or material/ parts information, and the PCB information read by the barcode reader on the machine or the conveyor during the assembly process.

Main function[*1]

(1) Defines/ read PCB barcodes
(2) Stores PCB information (model code, serial No.) read
(3) Stores trace data (machine transit time) of the machine
(4) Links PCB information to trace data
(5) Links PCB information to parts data (material verification)
(6) Generates / displays trace reports



*1 For using a traceability module, a material verification module (license) is also required.
*2 When placing a conveyor (middle conveyor) between two pieces of sequentially-positioned equipment, a PCB barcode reader and LMNC may be required for the conveyor.

Traceability level

・PCB trace
Specification of a PCB allows parts used with the PCB to be identified.

・Parts trace
Specification of a part allows identifying when and where the part was used.
Part-level traceability will be used together with the material verification function. In the material verification function, details on the installation time and attribute of a material mounted to each machine can be recorded. Such information and PCB data can be used, together with data including Part No., Lot, Vendor, Quantity, and User-defined item, to identify a PCB fabricated using the material.

・Placement point trace (for NPM X/NPM/AM series only)
On top of data recorded by using part-level traceability, placement point-level traceability can record even the reference number per placement point set by a data creation device. That allows identifying not only a PCB fabricated using the material based on data such as Part No., Lot, Vendor, Quantity, and User-defined item, but also the placement point.

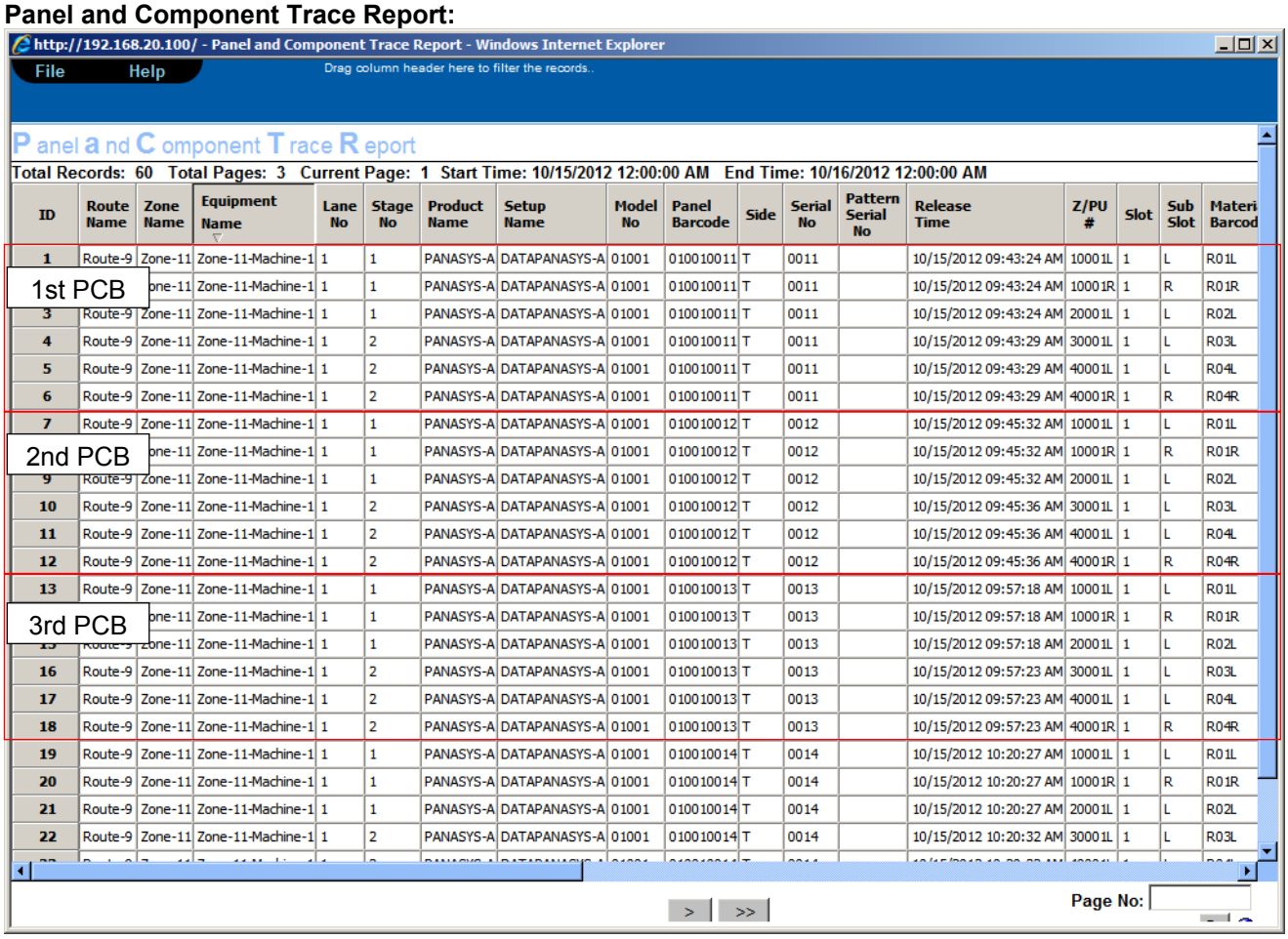
Function of Traceability Module

Main function	Detailed function		Description
	Large category	Small category	
Barcode reading function	PCB information	LMNC (Required for CM- / SP-based machine)	LMNC-ready installed barcode reader allows reading the following barcode formats (the head camera or installed scanner is used for NPM X/NPM/ AM series, SPV, SPG, and SPD) 1D code (Barcode): Code 39, Code 128, Code 93, Code 2of5, Interleaved, UPC, EAN, CODEBAR, MSI 2D code: Data Matrix, QR Code, PDF417
		Barcode specification	Alterable according to the barcode format
Data collection function	Data type		Model code, serial No.
	Number of machines which can collect data		100 stages(per PanaCIM controller server)
	Number of lines which can collect data		The equivalent of the number of stages above (per PanaCIM controller server)
	Collection interval		Every time a piece is produced
Production program check	Production program check		Urges a changeover when a PCB model code does not match production data being used for production by the mounter
Data storage function	Backup function	Automatic backup	Can back up data automatically at set intervals
		External storage	Backup data can also be stored in external media (other)
	Restoration function	Restore	Data can be restored based on backup data
Display function	Report display	Type	Reports shown in the next item can be displayed
		Save	Parameters (attribute value, range, etc.) necessary for preparing reports can be saved as report configuration files
		Quick-run	A report can be displayed by calling up the report configuration file saved
		Display time	After report settings are configured, reports can be displayed in a short time by carrying out report generation or quick run
Setting function	Trace level		Can define whether to conduct trace or not and, if conducted, the trace level
	Barcode settings		Can change PCB barcode settings (product code and serial No. reading)

Traceability report function

Traceability report	
Barcode trace	Can display reports based on all barcode data included in PCB information and parts information
Traceability of parts from PCB	Can display an information report about the parts used in production of the applicable PCB, based on PCB information
Traceability of PCB from parts	Can display an information report about the PCB produced using the applicable parts, the name of the line involved, the machine involved and the machine transit time, based on parts information
PCB trace	Can display a report about the name of the line where the applicable PCB was produced, the machine involved and the machine transit time

Example of a traceability report output screen
Panel and Component Trace Report:

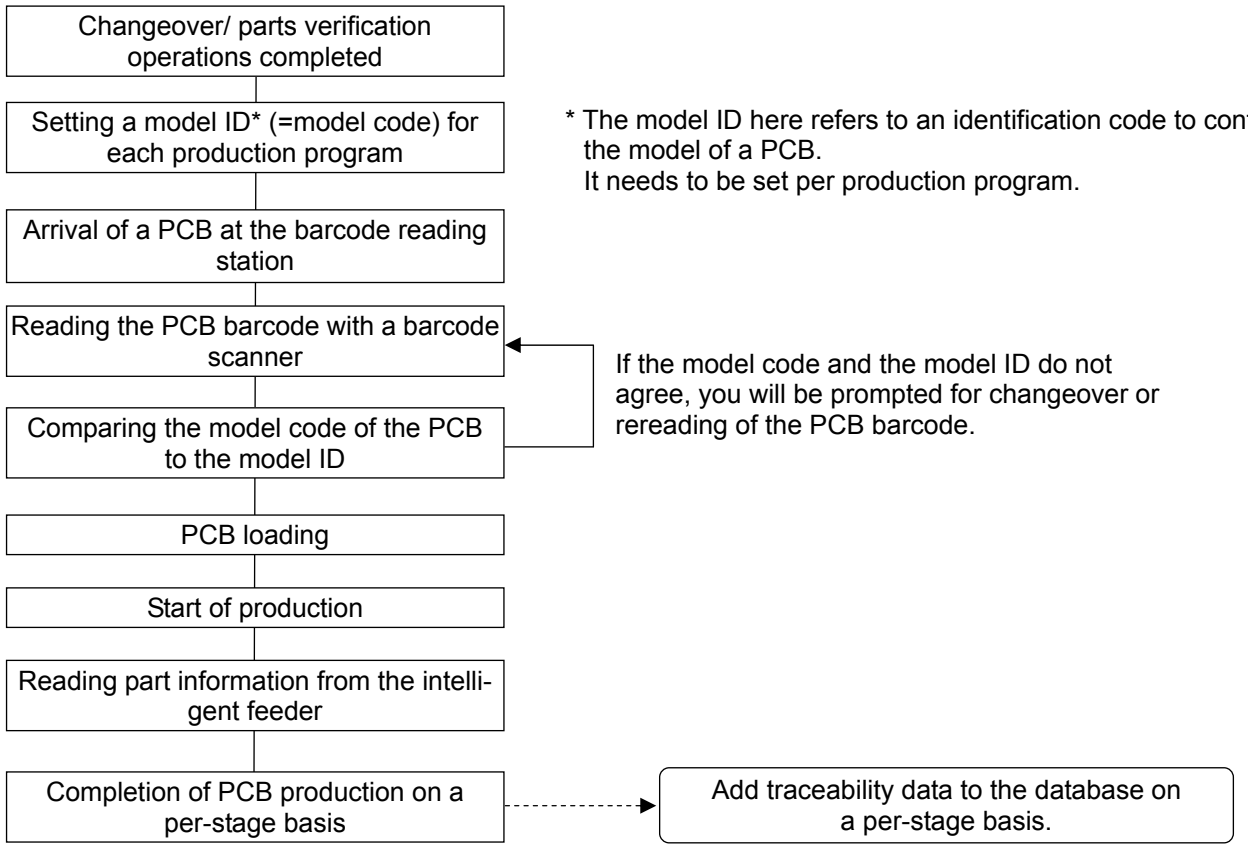


ID	Route Name	Zone Name	Equipment Name	Lane No	Stage No	Product Name	Setup Name	Model No	Panel Barcode	Side	Serial No	Pattern Serial No	Release Time	Z/PU #	Slot	Sub Slot	Materi Barcod
1	Route-9	Zone-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010011	T	0011		10/15/2012 09:43:24 AM	10001L	1	L	R01L
		one-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010011	T	0011		10/15/2012 09:43:24 AM	10001R	1	R	R01R
3	Route-9	Zone-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010011	T	0011		10/15/2012 09:43:24 AM	20001L	1	L	R02L
4	Route-9	Zone-11	Zone-11-Machine-1	1	2	PANASYS-A	DATAPANASYS-A	01001	010010011	T	0011		10/15/2012 09:43:29 AM	30001L	1	L	R03L
5	Route-9	Zone-11	Zone-11-Machine-1	1	2	PANASYS-A	DATAPANASYS-A	01001	010010011	T	0011		10/15/2012 09:43:29 AM	40001L	1	L	R04L
6	Route-9	Zone-11	Zone-11-Machine-1	1	2	PANASYS-A	DATAPANASYS-A	01001	010010011	T	0011		10/15/2012 09:43:29 AM	40001R	1	R	R04R
7	Route-9	Zone-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010012	T	0012		10/15/2012 09:45:32 AM	10001L	1	L	R01L
		one-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010012	T	0012		10/15/2012 09:45:32 AM	10001R	1	R	R01R
9	Route-9	Zone-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010012	T	0012		10/15/2012 09:45:32 AM	20001L	1	L	R02L
10	Route-9	Zone-11	Zone-11-Machine-1	1	2	PANASYS-A	DATAPANASYS-A	01001	010010012	T	0012		10/15/2012 09:45:36 AM	30001L	1	L	R03L
11	Route-9	Zone-11	Zone-11-Machine-1	1	2	PANASYS-A	DATAPANASYS-A	01001	010010012	T	0012		10/15/2012 09:45:36 AM	40001L	1	L	R04L
12	Route-9	Zone-11	Zone-11-Machine-1	1	2	PANASYS-A	DATAPANASYS-A	01001	010010012	T	0012		10/15/2012 09:45:36 AM	40001R	1	R	R04R
13	Route-9	Zone-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010013	T	0013		10/15/2012 09:57:18 AM	10001L	1	L	R01L
		one-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010013	T	0013		10/15/2012 09:57:18 AM	10001R	1	R	R01R
		one-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010013	T	0013		10/15/2012 09:57:18 AM	20001L	1	L	R02L
16	Route-9	Zone-11	Zone-11-Machine-1	1	2	PANASYS-A	DATAPANASYS-A	01001	010010013	T	0013		10/15/2012 09:57:23 AM	30001L	1	L	R03L
17	Route-9	Zone-11	Zone-11-Machine-1	1	2	PANASYS-A	DATAPANASYS-A	01001	010010013	T	0013		10/15/2012 09:57:23 AM	40001L	1	L	R04L
18	Route-9	Zone-11	Zone-11-Machine-1	1	2	PANASYS-A	DATAPANASYS-A	01001	010010013	T	0013		10/15/2012 09:57:23 AM	40001R	1	R	R04R
19	Route-9	Zone-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010014	T	0014		10/15/2012 10:20:27 AM	10001L	1	L	R01L
20	Route-9	Zone-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010014	T	0014		10/15/2012 10:20:27 AM	10001R	1	R	R01R
21	Route-9	Zone-11	Zone-11-Machine-1	1	1	PANASYS-A	DATAPANASYS-A	01001	010010014	T	0014		10/15/2012 10:20:27 AM	20001L	1	L	R02L
22	Route-9	Zone-11	Zone-11-Machine-1	1	2	PANASYS-A	DATAPANASYS-A	01001	010010014	T	0014		10/15/2012 10:20:32 AM	30001L	1	L	R03L

Output item list:

Output item	Description
Product name	The name of the program used for production
Setup Name	The name of MIX setup for common configuration
Model ID	Model code
Panel Barcode	Barcode information read from a PCB
Side	Front/ Rear
Serial No	Serial No. of a PCB read from a barcode
Pattern Serial No	(Not yet determined)
Release Time	Time spent unloading a PCB from production equipment
Z/ PU #	Table/ Slot/ Sub Slot numbers
Slot	Feeder slot number
Sub Slot	The right and left of the double feeder
Material ID	A unique identifier assigned per reel/ tray
Part Number	The name of parts verified
Expected Part	Displays the name of a legitimate part on production program as an alternate part is in use
Lot No	The production lot of a part
Vendor	The code of the manufacturer of a part
Feeder	Feeder serial No.
Quantity	The number of parts
Material Mount/Unmount Time	Time spent installing/ uninstalling a part
Operator ID	ID of an operator who performed verification
Order	Displays an operating instructions No. if registered in advance
Part Class	Rank: Mainly shows the LED brightness
MCID	A unique identifier correlated with a part No.
User Data 1 to 5	Individual user data (up to 5 items)

Traceability data generation flowchart

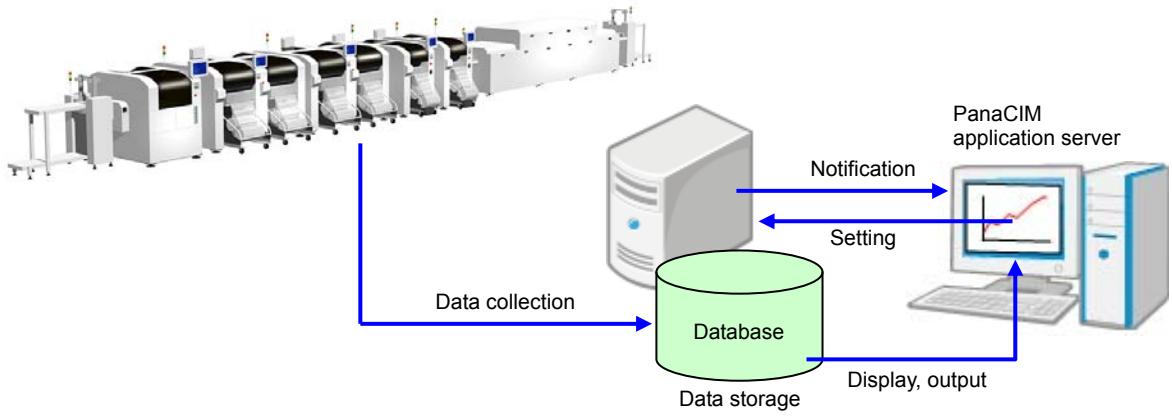


4.3 Production Analysis Module

A support function that, for the purpose of enhancing productivity (improved recognition/ pick-up error rate, reduced downtime, etc.) in the assembly line, collects operating information of the machine and then analyzes/ provides statistics of/ displays in real time the information.

Main function

(1) Collects operating performance information
(2) Displays the comparison of the operating performance and plan
(3) Graphically displays the operating performance and plan
(4) Analyzes operating performance



Function of Production Analysis Module

Main function	Detailed function		Description
	Large category	Small category	
Data collection function	Data type		Production control information that equipment outputs
	Number of machines which can collect data		100 stages (per PanaCIM controller server) * For express package: Up to 5 stages
	Number of lines which can collect data		The equivalent of the number of the stages above (per PanaCIM controller server) * For express package: Per line
	Collection interval		Per production PCB
Data storage function	Backup function	Automatic backup	Data can be backed up automatically at set intervals * For express package: Data retention period 30 days
		External storage	Backup data can also be stored in external media (other media)
	Restoration function	Restore	Data can be restored based on backup data
Display function	statistical graph display function	Type	Can display the operation analysis graph mentioned in the next item (partial)
		Save	・Parameters (attribute value, range, etc.) necessary for preparing graphs can be saved as graph configuration files ・Saves a created graph in image form (JPEG format)
		Quick-run	Calls up and graphically displays a graph configuration file saved
		Display time	After graph settings are configured, graphs can be displayed in short time by carrying out graph generation or quick run
	Report display	Type	Can display the operation analysis report described in the one after the one after the next item (partial)
		Save	Parameters (attribute value, range, etc.) necessary for preparing reports can be saved as report configuration files
		Quick-run	Displays a report by calling up the report configuration file saved
		Display time	After report settings are configured, reports can be displayed in a short time by carrying out report generation or quick run
	Real-time display	Automatic update	All graphs shown in the statistical graph display function can be updated automatically at specified intervals
		Update interval	Updated by minutes
		Save	Automatically saves an updated graph in image form (JPEG format)
Setup function	Graph display setting		Can configure settings (range, etc.) necessary for preparing graphs
	Threshold		Can set the threshold with respect to the graph attribute
	Threshold over notice	Highlighting	Notification will be visually provided when the graph attribute should exceed the set threshold

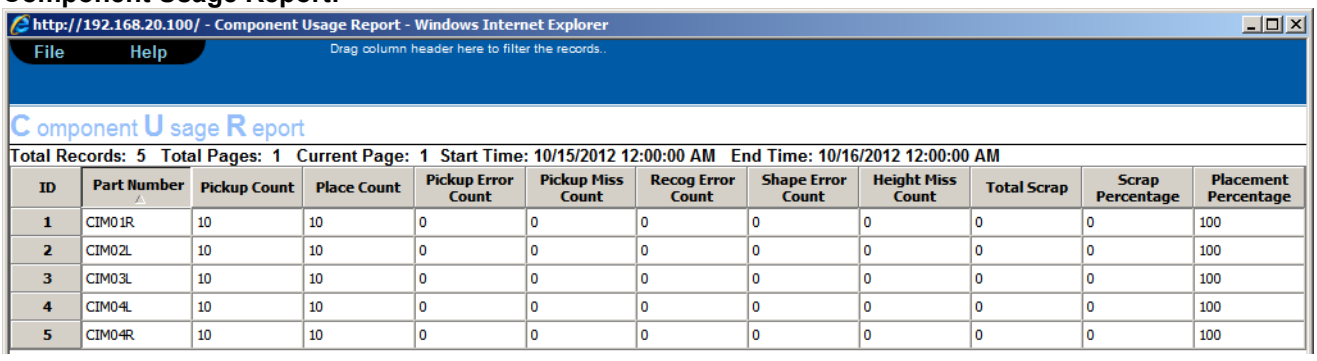
Production analysis graph (Major details)

Ratio analysis graph		
	Machine operation percentage	Displays model-/ PCB-specific machine operation rate (cumulative / per unit time) in a specified period of time
	Machine production percentage	Displays model-/ PCB-specific machine production rate (cumulative / per unit time) in a specified period of time
	Downtime ratio	Displays model-/ PCB-specific downtime rate operation rate (cumulative / per unit time) in a specified period of time
	Placement percentage	Displays model-/ PCB specific machine placement rate (cumulative / per unit time) in a specified period of time
Total error count analysis		
	Total error count analysis	Displays all model-/ PCB-specific error stop causes for a specified period of time
Standby count		
	Downtime count	Displays model-/ PCB-specific downtime count for a specified period of time
	Pre-process wait count	Displays model-/ PCB-specific pre-process wait count for a specified period of time
	Post-process wait count	Displays model-/ PCB-specific post-process wait count for a specified period of time
	Single/Cycle stop count	Displays model-/ PCB-specific single/cycle wait stop count for a specified period of time
	Part supply wait time	Displays model-/ PCB-specific part supply wait time for a specified period of time
	Solder paste supply wait count (Counter)	Displays model-/ PCB-specific solder paste supply wait count (counter) for a specified period of time
	Solder paste supply wait count (Sensor)	Displays model-/ PCB-specific solder paste supply wait count (sensor) for a specified period of time
	Cleaning paper supply wait count	Displays model /PCB specific cleaning paper supply wait count for a specified period of time
Machine status analysis		Reports model-/ PCB-specific machine status for a specified period of time

Production analysis graph (Major details)

Production model information report		
	Feeder arrangement	Feeder arrangement report by the specified production model
	Part arrangement	Part arrangement report by the specified production model
	Operating instruction history	Operating instruction history report by the specified production model
Operating analysis report		
	PCB production	Model-/ PCB-specific PCB production info report for a specified period of time
	Event trace	Model-/ PCB-specific event trace for a specified period of time
	Operating instructions in use	Model-/ PCB-specific operating instructions in use information report for a specified period of time
	Downtime	Model-/ PCB-specific downtime information report for a specified period of time
	Status trace	Model-/ PCB-specific status trace information report for a specified period of time
Material and parts verification report		
	Component Usage	Displays a list of all parts picked up or placed. Also displays the difference between picked-up parts and placed parts in line, machine and model.

Component Usage Report:



ID	Part Number	Pickup Count	Place Count	Pickup Error Count	Pickup Miss Count	Recog Error Count	Shape Error Count	Height Miss Count	Total Scrap	Scrap Percentage	Placement Percentage
1	CIM01R	10	10	0	0	0	0	0	0	0	100
2	CIM02L	10	10	0	0	0	0	0	0	0	100
3	CIM03L	10	10	0	0	0	0	0	0	0	100
4	CIM04L	10	10	0	0	0	0	0	0	0	100
5	CIM04R	10	10	0	0	0	0	0	0	0	100

4.4 Material Control Module

A support function used to manage the number and location of parts or feeders in inventory in the production or preparation area and also to reserve parts. The moisture control function exercises expiration date control over moisture sensitive devices (MSD).
It allows you to monitor in real-time the warehousing or shipment and usage of materials, and, based on a production order (lot), to arrange / obtain materials you need when you need them, thereby helping you promote the efficiency of operations and achieve inventory reduction.

Main function[*1]

The material control module consists of 3 individual modules separated by function.
You may select and purchase the license of each individual module, which permits you to use its function as appropriate.

Basic function (License)

(1) Control of individual IDs by material (reel, tray, etc.)
(2) Monitoring of the remaining quantity
(3) Management of the warehousing / shipment and location of parts

Material allocation function (License)

(4) Creating production orders
(5) Reserving parts and preparing the list of missing materials
(6) Shipment order and procurement of production materials

Moisture control function (License)

(7) Expiration date control over moisture sensitive devices (MSD)

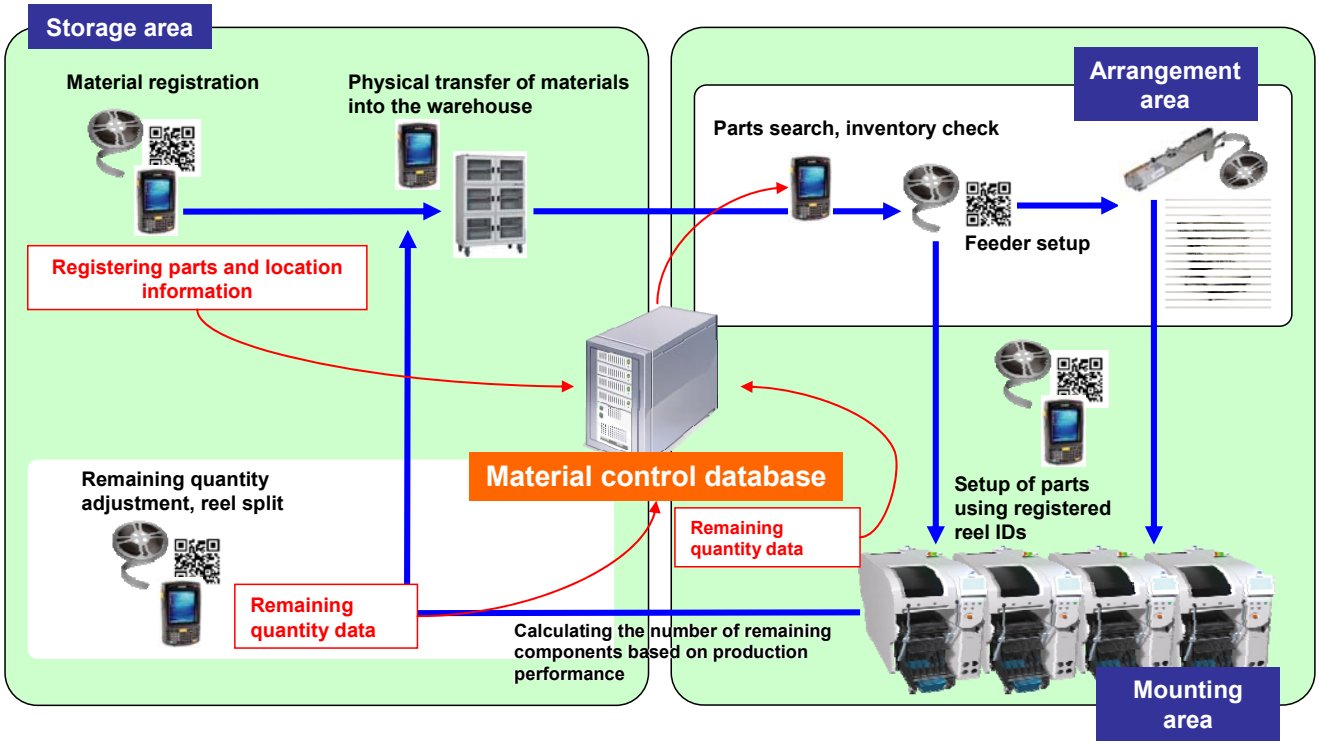
*1 For using a material control module, a material verification module (license) is also required.

Summary of material control module functions

Main function	Detailed function		Description	Required license*1		
	Large category	Small category		Basic	Material allocation	Moisture control
Material control basics	Remaining quantity control		Measures the remaining quantity based on production performance	○	–	–
	Warehouse in / out control		Allows controlling warehousing and shipment of materials based on material-specific IDs			
	Location control		Allows controlling material locations based on material-specific IDs			
Material allocation	Creation of production order (lot)		Allows arbitrary input of data about planned PCBs	○	○	–
	Material reservation, missing material list	Reserve	Reserves materials needed for each production order			
		Missing material list	Prepares a shortage list of materials that are required for each production order			
	Shipment order, picking	Shipment order	Prepares a list of materials needed for each production order, and allows it to be displayed with the material control scanner			
		Picking	Uses the material control wireless scanner to collect materials, which helps you understand the preparation state in real time			
Moisture control	MSD expiration date control		Controls the expiration date of moisture sensitive devices (MSD) ・ Stops the time counter in the dry box ・ Resets the time counter in the drying oven	○	–	○

*1 A Material Verification Module is also required (license).

Overview of basic / material allocation functions



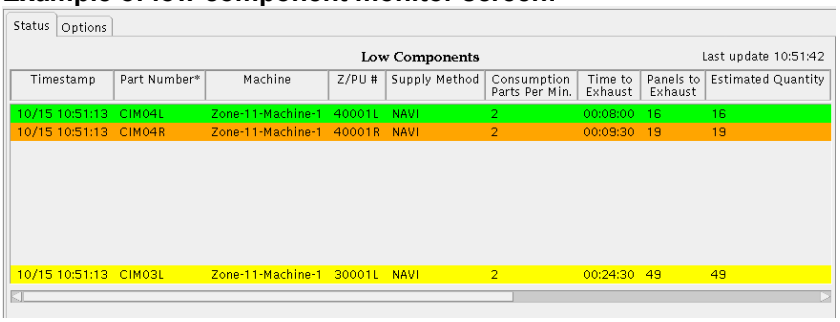
Summary of material control / basic module functions

Main function	Detailed function		Description
	Large category	Small category	
Remaining quantity control	Remaining quantity measurement		Measures the number of remaining reels/ trays on a per-material ID basis
	Remaining quantity monitor		Allows monitoring in real time the number of remaining parts being used in production and their usage conditions The display is color-coded according to warning level (floor by floor)
Warehouse in/ out and location control	Control of parts to be warehoused / shipped (Scanner operation)	Check-in	Scans and registers initial information, such as storage site, material ID, part name, Lot. Vendor and quantity, when warehousing new materials The locations of materials already registered can be controlled/ changed by a means of scanning of their storage site and material IDs
		Remaining quantity adjustment	Controls the number of remaining parts after new parts are registered, or the remaining number of parts being in use for production is adjusted
		Auto remaining quantity clear	Automatically clears to 0 the number of target parts remaining in the event of parts shortages or at the time of detection of the splicing tape
		Manual remaining quantity clear	Allows clearing target parts to 0 manually
		Part split	Splits registered parts and allows materials to be used and controlled individually
	Inventory/ location control (Scanner/ Web application operation)	Material status check	Displays information on the details of material IDs registered (storage site, quantity, etc.)
		Inventory search	Part search: Enters the name of a targeted part to display a list of the locations of the part Feeder search: Enter the width / pitch of a targeted feeder to display a list of the locations of the feeder
		Storage unit Inventory control	Displays a list of materials that are kept in the storage unit you inputted
		Storage unit Transfer	Changing the location of the storage unit with materials included allows controlling/ changing of the locations of all the materials in the storage unit (Scanner operation only)
	Production (Scanner operation)	Material ID verification	Parts verification can be completed at the time of production just by scanning the material ID; however, provided that the material ID needs to have been checked in
		Restrictive use of old reels	Prohibits those parts, the remaining number of which is cleared to 0, from being reserved or verified
		Material ID rereading prevention	Prohibits the barcode of a part being used to be used for verifying a new part when the new part is installed, thereby vigorously preventing erroneous production

Summary of material control / basic module functions (Continued)

Main function	Detailed function		Description
	Large category	Small category	
Production order control	New creation	Manual creation	Allows scheduling production orders according to production schedule
		CSV import	Allows importing the production orders from CSV
	Input item		Production order ID, Route (Production line), PCB name, Placement side, Mix name, Tact time, Patterns, Production patterns, Start and end time, Changeover time * Some items are set automatically based on a production program
Report	Inventory /Location control		Allows displaying a report of part inventory and location lists
	Material quantity report		Allows displaying a report about the number of parts remaining
	Material quantity adjust report		Allows displaying a history of changes made to the number of parts remaining
	Used material report		Allows displaying a list of parts used
	Material location history report		Allows displaying a history of the locations of parts used
	Feeder position history report		Allows displaying information on the position of feeders, provided that the feeder control is exercised
	Material move report		Allows displaying a report about a moving history of materials

Example of low component monitor screen:



→ Replenishment part already prepared: Green

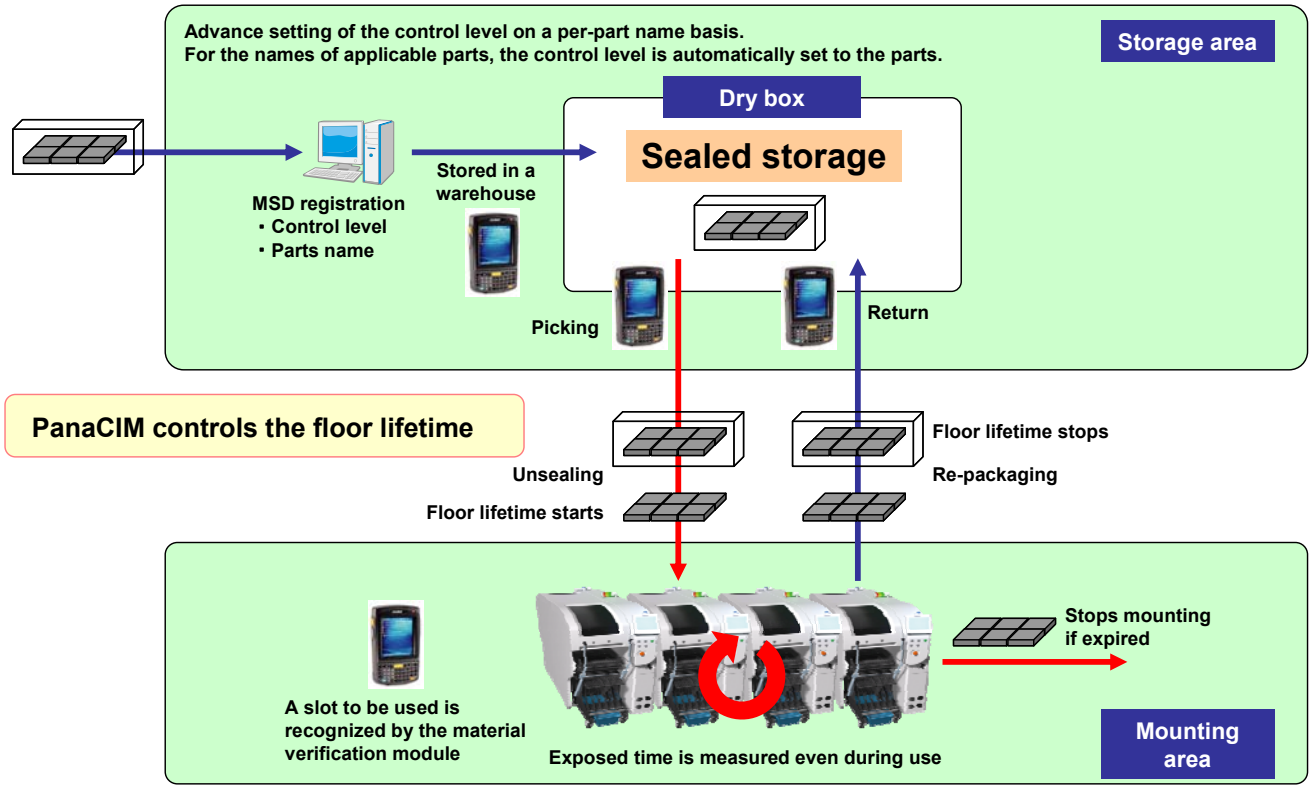
→ Warning level 2: Orange

→ Warning level 1: Yellow

Summary of material control / material allocation module functions

Main function	Detailed function		Description
Production order control (Web application operation)	New creation	Manual creation	Enables to schedule production orders according to production schedule
		CSV import	Enables to import the production orders from CSV
		Master production order	Enables to create a production order linked to a master (parent) production order
	Input item		Production order ID, Route (Production line), PCB name, Placement side, Mix name, Tact time, Patterns, Production patterns, Start and end time, Changeover time * Some items are set automatically based on a production program
Part reserve and shortage list	Material reserve (Web application operation)	Reserve	Reserves allocation of materials necessary for a production order created
		Automatic allocation	Allows automatically allocating materials within a designated area Manual modification is also allowed to designate a specified material
		Missing material list	Allows displaying a list of missing materials that are required for production, which helps prompt the missing parts to be added to the warehouse
		Status display	Displays the status of each production order that is created new, already reserved or already collected by using icons, or by a means of color coding
	Material replenishment (Scanner operation)	Replenishment of missing material	Registers and adds materials as soon as the deficiency of the materials is supplied
Shipment order and procurement	Shipment order (Scanner/ Web application operation)	Material list display	Prints out or displays with a scanner a list of the IDs and locations of materials used for a production order
		Batch list display	Allows collectively displaying lists of materials used for a production order linked to a master (parent) production order
	Procurement (Scanner operation)	Pickup	Picks up necessary materials, listing or scanning the materials used for a production order
		Batch pickup	Allows a collective pick-up of materials used for a production order linked to a master (parent) production order

Overview of moisture control function



Detailed function of material control / Moisture control modules

Main function	Detailed function		Description
	Large category	Small category	
Lifetime control	Shelf lifetime	Time control of unopened parts	Manages time in such a way that the duration of time during which a moisture sensitive device package is unopened is counted as a shelf time (a maximum of 12 months)
	Floor lifetime	Time control of opened parts	Manages time in such a way that the duration of time during which a moisture sensitive device package is opened is counted as a floor lifetime (Followed by a moisture sensitive level)
MSD control	Warehouse in/out control (Scanner operation)	MSD registration	Additionally registers a moisture sensitive level, as well as a normal material ID and remaining quantity, when the moisture sensitive devices are stored in a warehouse.
		Unsealing treatment	Performs unsealing treatment when you unseal a package of moisture sensitive devices, and then starts counting down the floor lifetime
		Sealing treatment	Performs sealing treatment when you seal a package of the moisture sensitive devices, and then stops counting down the floor (following the moisture sensitive device level)
		Removal from dry box	Performs a removal procedure when you take moisture sensitive devices out of the dry box, and then starts counting down the floor lifetime
		Return to dry box	Performs a warehousing procedure when you return the moisture sensitive devices to the dry box, and then stops counting down the floor lifetime (following the moisture control part level)
	Status control (Web application operation)	MSD inventory control	On top of the remaining quantity/ location, attribute information (level, expiration date) of moisture sensitive devices can be controlled
		MSD status monitor	The status (floor lifetime) of moisture sensitive devices can be displayed in real time. Displays it in real time according to the used hours or warning settings.
	Production time control	Limitations on use of expired MSD (for material verification)	Returns an error when expired moisture sensitive devices were verified, and disallows the use of the expired moisture sensitive parts
		Interlock for expired MSD (During production)	Temporarily stops production when a moisture sensitive device being used in the production expires, and resume the production only after a new part (before the expiration date) is verified
Report	MSD status report		Displays a report of moisture sensitive device status
	MSD use history report		Displays a history of moisture sensitive device

Moisture sensitive device level and expiration data
Based on the moisture sensitive level (MSD) described in the part package, the floor lifetime (expiration date) is controlled.

Moisture sensitive level (MSL): Compliance with IPC/JEDEC J-STD-033B

Moisture sensitive level	Storage conditions of unsealed parts	
	Floor lifetime	Environment
1*1	Unlimited	30 °C or less, Humidity 85 % or less
2	1 year	30 °C or less, Humidity 60 % or less, or specifically defined value
2a	4 weeks	
3	168 h	
4	72 h*2	
5	48 h*2	
5a	24 h*2	
6*1	Must dry before use. Reflow within a specified period of time specified on the label after drying.	

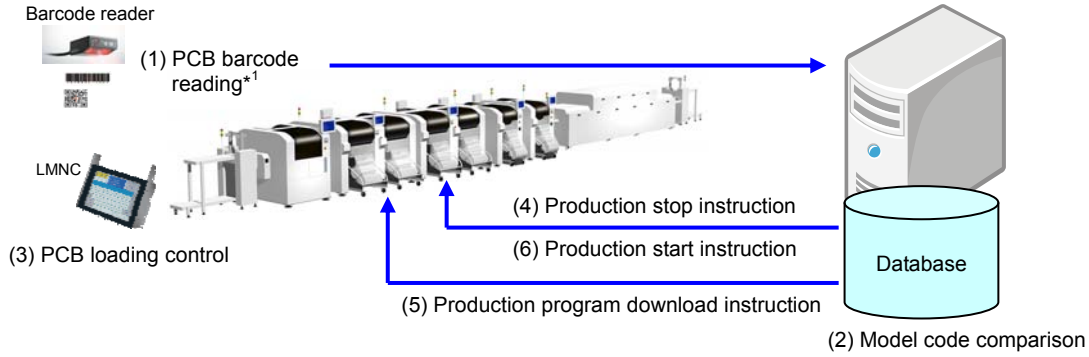
*1 Levels 1 and 6 are not supported.
*2 Moisture sensitive device standard: Based on IPC/JEDEC J-STD-033B, the floor lifetime of a part of which the moisture sensitive level is 4 or higher cannot be stopped.
Additionally, drying treatment (baking treatment) cannot be used to reset the floor lifetime.

4.5 Production Changeover & Control Module

A support function intended to reduce product changeover time through automatically downloading a production program and also automatically adjusting the lane width according to the PCB ID.
Production changeover and control of the NPM X/NPM/AM series, SPV, SPG, SPD, and SP70 (LNB connection) will be performed and controlled (external scanner read type, head read type, planning report read type) by the machine.
(For details, please refer to the “Specification” booklet of each model.)

Main function

(1) PCB barcode definition and reading
(2) Model code comparison
(3) PCB loading control
(4) Machine control (Production stop instruction)
(5) Machine control (Production program download instruction)
(6) Machine control (Production start instruction)



*1 When placing a conveyor (middle conveyor) between two pieces of sequentially-positioned equipment, a PCB barcode reader and LMNC may be required for the conveyor.

List of production changeover and control functions

Main function	Detailed function		Description
	Large category	Small category	
Barcode definition and reading function	PCB information	Format	1D code (Barcode): Code 39, Code 128, Code 93, Code 2of5, Interleaved, UPC, EAN, CODEBAR, MSI 2D code: Data Matrix, QR Code, PDF417
Model code comparison	Model code registration		Links the model code on a barcode to a production program name before start of production
	Model code comparison		Compares the model code of the PCB read to the one being used in production, and determines whether to do a changeover
Machine control	Production stop		A changeover is considered necessary when the model code of the PCB read does not match the one being used in production, and the production will be automatically suspended after all PCBs in progress are ejected from the machines in sequential order starting with the fist machine in the line
	Production program download		Automatically downloads a production program to the machines under suspension in sequential order starting with the first one in the line
	Production start		Automatically makes the machines, which completed the download of the production program, start production in sequential order starting with the first one in the line
Error handling	-		Displays an error message on LMNC when an error occurs in reading, or when a model code yet to be registered is read

Production changeover and control report function

Production changeover and control report	
Product report	Displays a changeover summary report

4.6 Production Monitoring & Dispatch Module

A support function that obtains production/ operating information in real time and displays the condition of the line on the monitor or PDA, which provides to the operator instructions on what to do in the event of an error or parts shortages and thus helps enhance productivity, thereby reducing the downtime of the assembly line or allowing for advance preparation of parts.
The Component Supply Navigator (option) optimizes the supply sequence to provide a shortest route without component shortage. It can prevent short time breakdown in the production line and contribute to manpower saving.

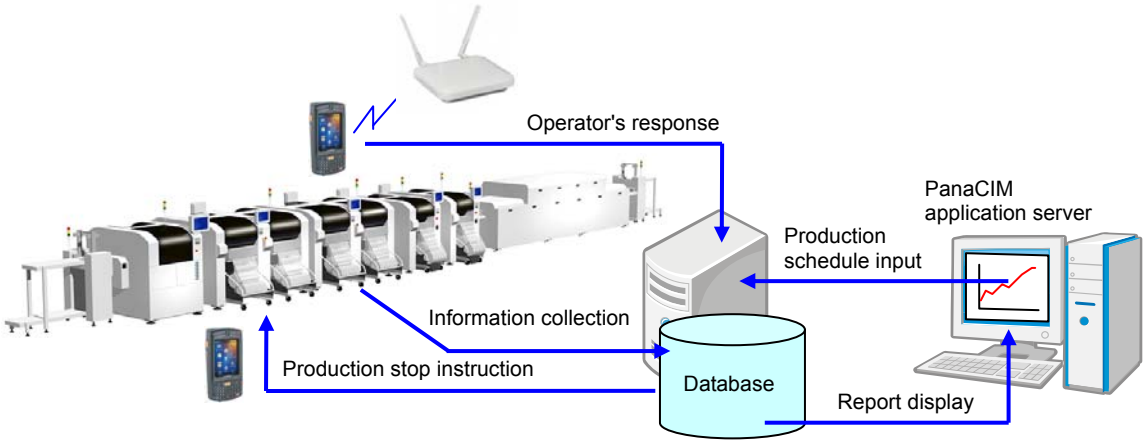
* Please install a Production Monitoring Client if the Production Monitoring Module is used and the number of stages is more than 7 per line and the number of lines is more than 4.
The necessary number of pieces of equipment depends on the machine configuration. So, for details, please contact us.

Main function[*1]

(1) Monitors line in an entire factory
(2) Collects real time event and displays warning as necessary
(3) Inputs production
(4) Displays changeover information
(5) Displays an operator's response status

Option function（Component Supply Navigator）[*2]

(6) Optimizes the supply sequence to provide a shortest route without component shortage.
(7) Displays the best supply route on wireless scanner for each operator.
(8) Optimizes supply sequence of multiple lines.



*1 If you use a production monitor & dispatch module, a material verification module (license) is also required.
To display production information of the machine, a production analysis module is required.
*2 Component Supply Navigator (license) is required.

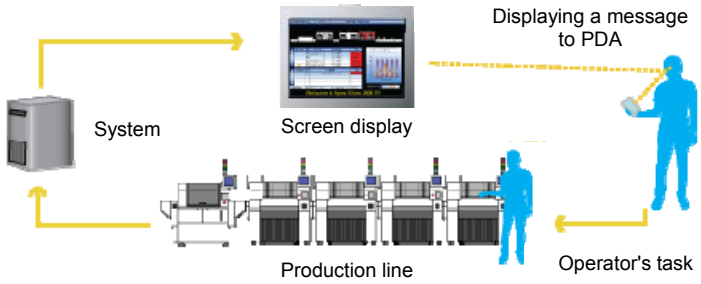
Function of Production Monitoring & Dispatch Module

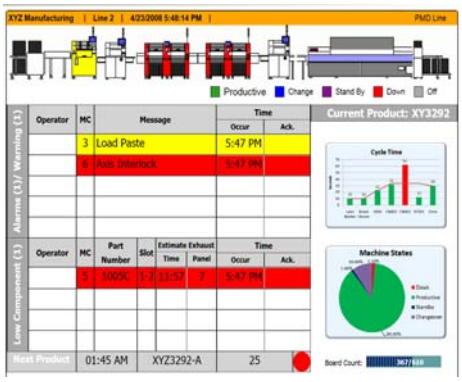
Main function	Detailed function		Description
	Large category	Small category	
Event monitor	Production Information/ Real time event monitor	Monitor item/ Display information	Displays machine status, real time event, production analysis chart, etc.
		Real time event display	Displays machine event, part verification event
			Error events are displayed at the top
		Event type	Part exhaust warning, part exhaust stop error, machine stop error, etc.
	Screen customization	Appearance	The appearance of the monitor can be customized
		Display target	Cycle time, Production percentage, Pickup percentage, Produced panels, Actual cycle time, Production Analysis Chart, Production Analysis report and Machine event
	Monitor device		Allows displaying on the monitor using a VGA-ready device
Custom message	Display to monitor		Displays a custom message to any line at a specified intervals
Check function (Scanner operation)	Setup	Line selection	Designates a line that receives an event (multiple selections)
		Event selection	Can select events to receive ((from among) equipment error events only, parts error events only, both)
	Operation	Event check reply	Acknowledges events on a scanner
		Event check reply cancel	Cancels a reply you acknowledged once on a scanner
		Event forced completion	Allows forcefully setting an event up as completed on a scanner (select a reason for the execution of forced completion)
	Recode	Event check reply record	Records the ID of the operator who acknowledged an event check and the time when the check was acknowledged
		Event check reply cancel record	Records the ID of the operator who cancelled an event check reply and the time when the reply was cancelled
		Event forced completion record	Records the ID of the operator who forcefully completed an event and the time when the event was forcefully completed
Production order interlock	Display of schedule		Displays a production schedule based on operating instructions
	Units for displaying schedule		Production schedules can be expressed on a day-to-day or week-to-week basis
	Production order control		Enables to add, edit, delete and import a production order
			Items to be entered when additional instructions are provided: Production order name, Route name, Production program name, Placement side, Mix name, Line tact time, Patterns/panel, Production scheduled patterns, Production start time, Production end time, Changeover time, etc.
Changeover information display	Settings	Line selection	Allows setting of a line the changeover information of which is displayed on the monitor
		Notification timing selection	Allows setting of the timing for displaying changeover information
			Remaining production time: Notification of the remaining time before the completion of current production
	Display		Displays the name of the next program on the monitor, based on the operating instructions entered in a schedule

Production monitor & dispatch report function

Production monitor & dispatch report	
PMD event history	Displays event history
Instruction to Operator	Displays instructions to the operator
Materials-in-process	Displays items in process
Next task instruction	Instructs next task to the operator
Downtime	Displays machine downtime

Example of a production monitor & dispatch screen





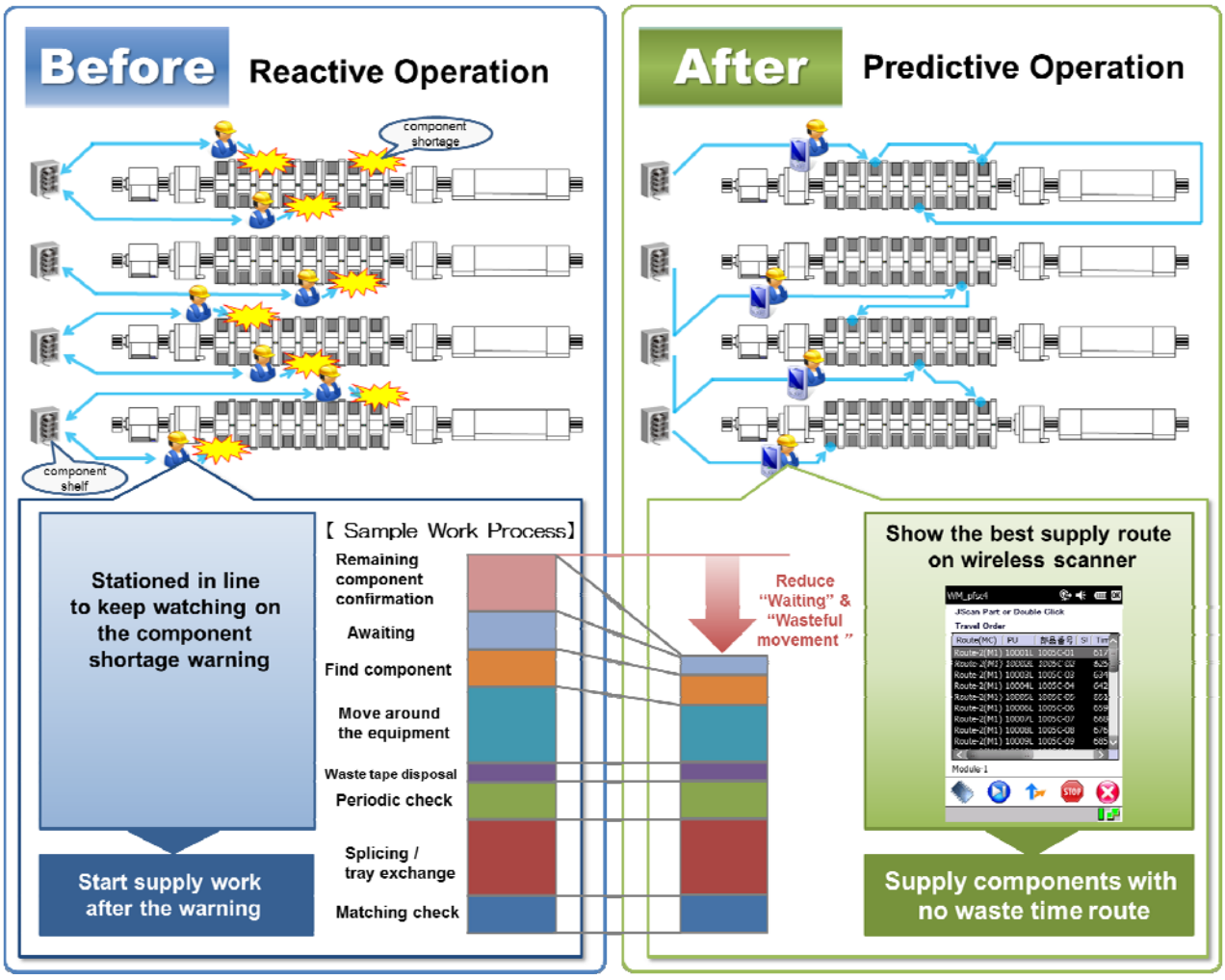
■ Component Supply Navigator Function

Main function	Detailed function	Description
Component Supply Navigator (Option)	Supply sequence optimize	Optimizes the supply sequence to provide a shortest route without component shortage.
	Supply route instruction generate	Generates the best supply route instruction and sends it to each operator.
	Supply route instruction UI	Displays a component supply list (supply rote list) with the best supply sequence on wireless scanner for each operator
	Work progress monitor	Displays the component supply list and work progress of each operator in real time.
	Component location display※1	Displays component locations in the component supply rote instruction list.
	Component replenishment instruction of component shelf in the line※1	Displays a replenishment instruction on monitor when the components which required for production are not on the shelf in the line.
	Work performance control of each operator※2	Displays the work history and work efficiency of each operator.

※1 The basic function (license) and material allocation function (license) of Material Control Module are required.

※2 Production Analysis Module (license) is required.

■ Component Supply Navigator : Effect & Operation window



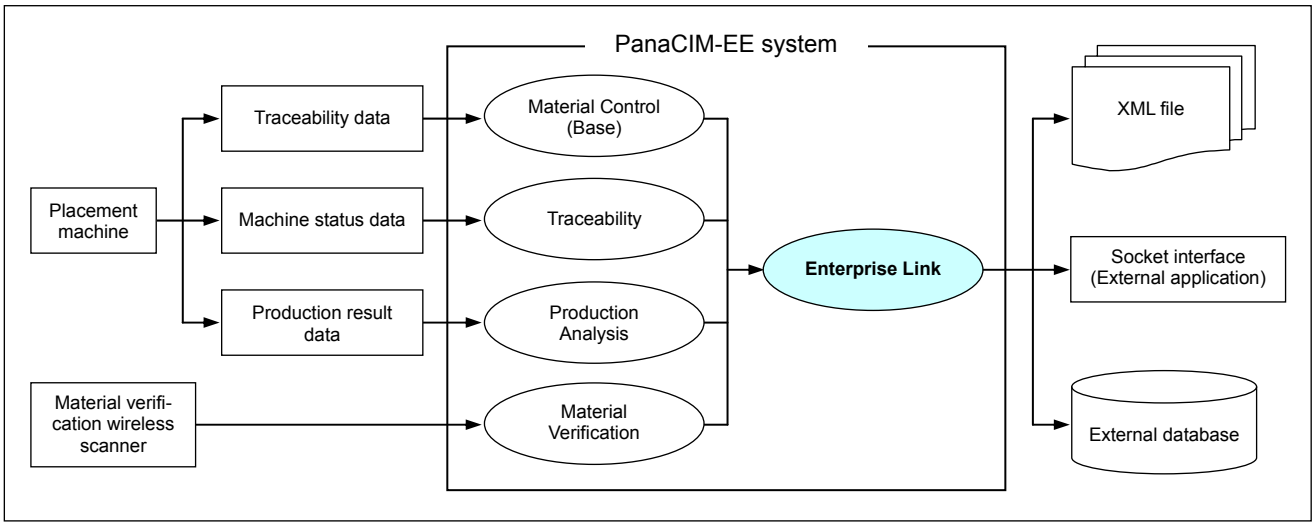
4.7 Enterprise Link Module

A function designed to output to external applications the data (material verification data, trace data, operating performance data) collected from the machine during production.

Main function

(1) Exports data using a XML file, socket interface, or database interface
* Exportable data: PCB trace data, material verification data, equipment status data, the number of PCBs produced

(2) Serves as a central repository relay server for data being exported, which transfers data, or processes data in the event of any error, and thus achieves reliable LAN asynchronous communication



Function of Enterprise Link Module

Main function	Detailed function		Description		
	Large category	Small category			
Export format	XML		XML interface		
	Socket		Socket interface		
	Database		Database interface		
Export data	Data category	Export format	XML file	Database (SQL)	Socket
	Traceability		○	○	○
	Machine status		○	−	−
	Material verification		○	−	−
	Scrap data		○	−	−
	Production PCB data		○	−	−

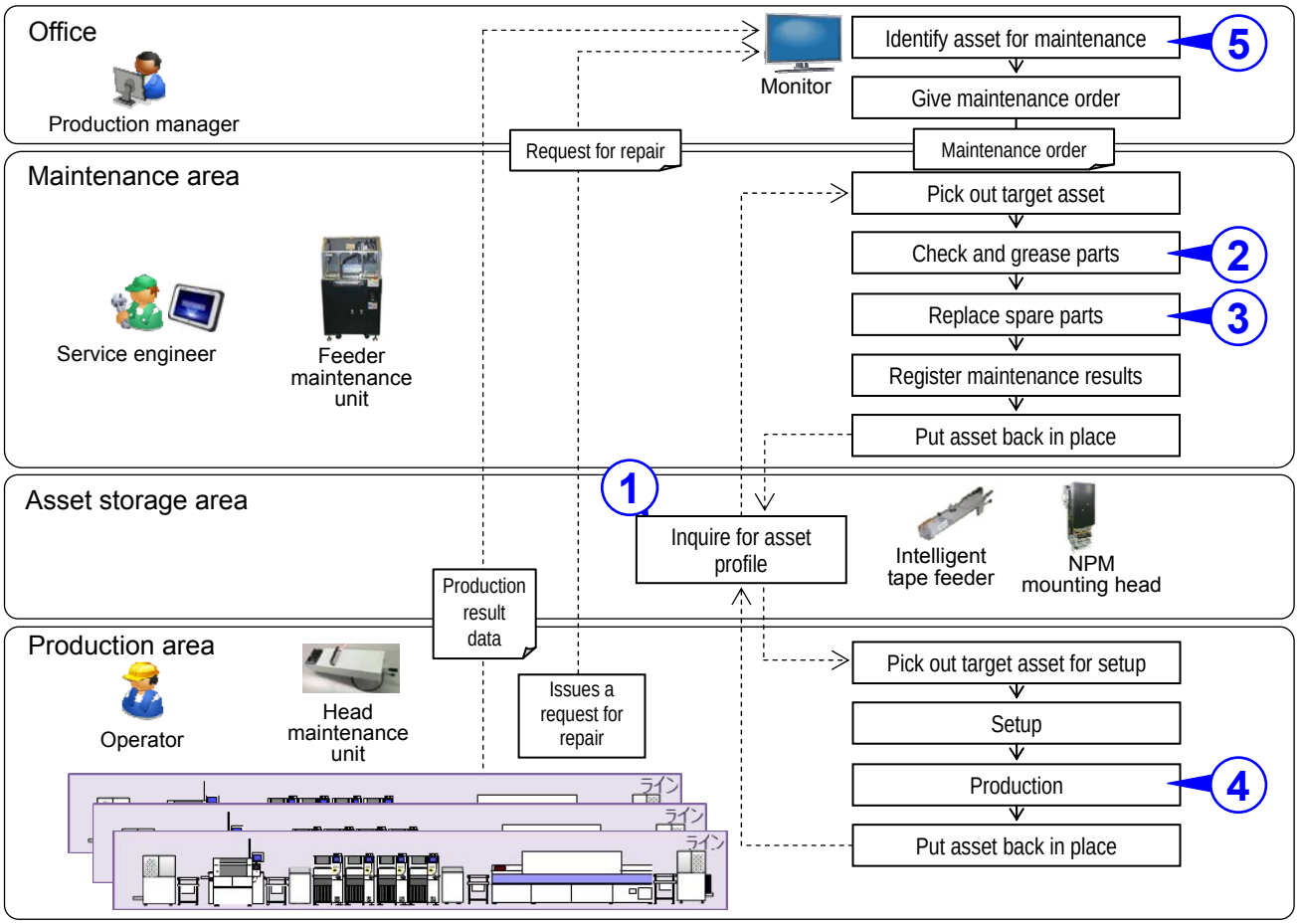
○: Applicable, −: Not applicable

4.8 Maintenance Module

A support function used to manage the profile of the assets (machines, heads, feeders, etc.) on the assembly floor and to record the history of maintenance per asset. This helps reduce the time required for maintenance planning and management operations.

■ Main function

The following shows the schematic of expected operation flow and main functions of this module.



No.	Main function	Description
1	Asset profile management	Centralizes the management of profile information (serial number, etc.) of assets with database. The assets include machines, feeders and heads that are used in the mounting processes.
2	Maintenance history management	Displays the maintenance items, asset by asset, online on the Android tablet, and records the history of maintenance performed.
3	Spare parts replacement history management	Displays the quantity of spare parts required for the maintenance, asset by asset, online on the Android tablet, and records the history of parts replaced.
4	Prevention of misuse of bad asset	Holds back production if the machine has been misloaded with any asset to which a maintenance order was given (locked on the system).
5	Maintenance information centralized monitor*1	Displays the assets that will need maintenance soon on the monitor.

*1 To use this function, Production Monitor & Dispatch module license is required.

■ Basic specification

Item		Description
Target assets	Equipment	Refer to “5.1”.
	Accessories	● Intelligent tape feeders ● Slim single feeders ● NPM mounting heads ● Mounting nozzles (with serial barcode)
Connectable maintenance units		● Feeder maintenance unit ● Head maintenance unit

5. Other

5.1 Target machines

Machines for Production Analysis, Material Verification, Traceability modules

No.	Model ID	Model No.	Production Analysis				Material Verification				Traceability			
		Function	PCB output volume	Production percentage	Wait stop time	Placement percentage count, etc.	Operator notification	Reel and tray parts	Solder paste, mask, squeegee	Machine interlock	PCB trace	Part trace	Placement point trace	Trace data control
1	NPM-WXS	NM-EJM2E	○	○	○	○	○	○	−	○	○	○	○	○
2	NPM-WX	NM-EJM9D	○	○	○	○	○	○	−	○	○	○	○	○
3	NPM-DX*3	NM-EJM8D	○	○	○	○	○	○	−	○	○	○	○	○
4	NPM-D3	NM-EJM6D	○	○	○	○	○	○	−	○	○	○	○	○
5	NPM-D2	NM-EJM5D	○	○	○	○	○	○	−	○	○	○	○	○
6	NPM-D	NM-EJM1D	○	○	○	○	○	○	−	○	○	○	○	○
7	NPM-TT2	NM-EJM1E	○	○	○	○	○	○	−	○	○	○	○	○
8	NPM-TT	NM-EJM3D	○	○	○	○	○	○	−	○	○	○	○	○
9	NPM-W2S	NM-EJM5E	○	○	○	○	○	○	−	○	○	○	○	○
10	NPM-W2	NM-EJM7D	○	○	○	○	○	○	−	○	○	○	○	○
11	NPM-W	NM-EJM2D	○	○	○	○	○	○	−	○	○	○	○	○
12	NPM	NM-EJM9B	○	○	○	○	○	○	−	○	○	○	○	○
13	AM100	NM-EJM4D	○	○	○	○	○	○	−	○	○	○	○	○
14	CM602-L	NM-EJM8A	○	○	○	○	○	○	−	○	○	○	×	○
15	CM232-M	NM-EJM9C	○	○	○	○	○	○	−	○	○	○	×	○
16	CM212-M	NM-EJM6A	○	○	○	○	○	○	−	○	○	○	×	○
17	CM101-D	NM-EJM5B	○	○	○	○	○	○	−	○	○	○	×	○
18	CM402-L	KXF-4Z4C	○	○	○	○	○	○	−	○	○	○	×	○
19	CM402-M	KXF-4Y3C	○	○	○	○	○	○	−	○	○	○	×	○
20	CM401-L	NM-EJM1A	○	○	○	○	○	○	−	○	○	○	×	○
21	CM401-M	KXF-4X3C	○	○	○	○	○	○	−	○	○	○	×	○
22	DT401-F	KXF-E64C	○	○	○	○	○	○	−	○	○	○	×	○
23	DT401-M	KXF-E53C	○	○	○	○	○	○	−	○	○	○	×	○

(○: Available, ×: Unsupported, −: Not applicable)

*1 Please make an inquiry about any other machines than those of the described above, separately.
*2 The marks at the top in the SP70 fields show the interface conditions that are available when it is connected to LNB, and, the marks at the bottom in the fields, the interface conditions that are available when it is connected to PT200.
*3 NPM-DX does not support tray parts.

Machines for Production Analysis, Material Verification, Traceability modules

No.	Model ID	Model No.	Production Analysis				Material Verification				Traceability			
		Function	PCB output volume	Production percentage	Wait stop time	Placement percentage count, etc.	Operator notification	Reel and tray parts	Solder paste, mask, squeegee	Machine interlock	PCB trace	Part trace	Placement point trace	Trace data control
24	SPV	NM-EJP7A NM-EJP8A	○	○	○	—	×	—	○	○	○	—	—	○
25	SPD	NM-EJP5A	○	○	○	—	×	—	○	○	○	—	—	○
26	SPG	NM-EJP6A	○	○	○	—	×	—	○	○	○	—	—	○
27	SP70*2	NM-EJP3A	×	×	×	—	×	—	○	○	×	—	—	×
			×	○	○	—	×	—	○	×	—			
28	SP60P-MU	KXF-1J3C	×	○	○	—	×	—	○	×	—			
29	SP60P-L	KXF-1G4C	×	○	○	—	×	—	○	×	—			
30	SP80V-MU	KXF-1K3C	×	○	○	—	×	—	○	×	—			
31	SP80V-L	KXF-1H4C	×	○	○	—	×	—	○	×	—			

(○: Available, ×: Unsupported, —: Not applicable)

*1 Please make an inquiry about any other machines than those of the described above, separately.
*2 The marks at the top in the SP70 fields show the interface conditions that are available when it is connected to LNB, and, the marks at the bottom in the fields, the interface conditions that are available when it is connected to PT200.
*3 NPM-DX does not support tray parts.

Machines for Material Control, Production Changeover & Control, Production Monitor & Dispatch modules

No.	Model ID	Model No.	Material Control				Production Changeover				Production Monitor				
		Function	Remaining quantity monitoring	Warehouse in / out Location control	Part reservation, List creation of part shortage	Expired data control of MSD	PCB barcode reading	Model code comperision	PCB loding control	Machine control	Line monitor in an entire factory	Collection and warning display of real time event	Changeover information display	Instructions to operator	Component supply navigator (Supply route instructions) *4
1	NPM-WXS	NM-EJM2E	○	○	○	○	—*2				○	○	○	○	○
2	NPM-WX	NM-EJM9D	○	○	○	○	—*2				○	○	○	○	○
3	NPM-DX	NM-EJM8D	○	○	○	○	—*2				○	○	○	○	○
4	NPM-D3	NM-EJM6D	○	○	○	○	—*2				○	○	○	○	○
5	NPM-D2	NM-EJM5D	○	○	○	○	—*2				○	○	○	○	○
6	NPM-D	NM-EJM1D	○	○	○	○	—*2				○	○	○	○	○
7	NPM-TT2	NM-EJM1E	○	○	○	○	—*2				○	○	○	○	○
8	NPM-TT	NM-EJM3D	○	○	○	○	—*2				○	○	○	○	○
9	NPM-W2S	NM-EJM5E	○	○	○	○	—*2				○	○	○	○	○
10	NPM-W2	NM-EJM7D	○	○	○	○	—*2				○	○	○	○	○
11	NPM-W	NM-EJM2D	○	○	○	○	—*2				○	○	○	○	○
12	NPM	NM-EJM9B	○	○	○	○	—*2				○	○	○	○	○
13	AM100	NM-EJM4D	○	○	○	○	—*2				○	○	○	○	○
14	CM602-L	NM-EJM8A	○	○	○	○	○	○	○	○	○	○	○	○	○
15	CM232-M	NM-EJM9C	○	○	○	○	○	○	○	○	○	○	○	○	○
16	CM212-M	NM-EJM6A	○	○	○	○	○	○	○	○	○	○	○	○	○
17	CM101-D	NM-EJM5B	○	○	○	○	○	○	○	○	○	○	○	○	○
18	CM402-L	KXF-4Z4C	○	○	○	○	○	○	○	○	○	○	○	○	○
19	CM402-M	KXF-4Y3C	○	○	○	○	○	○	○	○	○	○	○	○	○
20	CM401-L	NM-EJM1A	○	○	○	○	○	○	○	○	○	○	○	○	○
21	CM401-M	KXF-4X3C	○	○	○	○	○	○	○	○	○	○	○	○	○
22	DT401-F	KXF-E64C	○	○	○	○	○	○	○	○	○	○	○	○	○
23	DT401-M	KXF-E53C	○	○	○	○	○	○	○	○	○	○	○	○	○

(○: Available, ×: Unsupported, —: Not applicable)

*1 Please make an inquiry about any other machines than those of the described above, separately.

*2 NPM X/NPM/AM series, SPV, SPD, SPG and SP70 (LNB connection) support Production Changeover & Control by the machine side options.

*3 The marks at the top in the SP70 fields show the interface conditions that are available when it is connected to LNB, and, the marks at the bottom in the fields, the interface conditions that are available when it is connected to PT200.

*4 If using a mixed line of NPM X/NPM/AM series and CM series (CM602, CM101, CM232, CM212, CM402/ DT401), please contact us.

Machines for Material Control, Production Changeover & Control, Production Monitor & Dispatch modules

No.	Model ID	Model No.	Material Control				Production Changeover				Production Monitor				
		Function	Remaining quantity monitoring	Warehouse in / out Location control	Part reservation, List creation of part shortage	Expired data control of MSD	PCB barcode reading	Model code comperision	PCB loding control	Machine control	Line monitor in an entire factory	Collection and warning display of real time event	Changeover information display	Instructions to operator	Component supply navigator (Supply route instructions) *4
24	SPV	NM-EJP7A NM-EJP8A	—	○	○	—	—*2				×				
25	SPD	NM-EJP5A	—	○	○	—	—*2				×				
26	SPG	NM-EJP6A	—	○	○	—	—*2				×				
27	SP70*3	NM-EJP3A	—	×	×	—	—*2				×				
			—	×	×	—	○	○	○	○	×				
28	SP60P-MU	KXF-1J3C	—	×	×	—	○	○	○	○	×				
29	SP60P-L	KXF-1G4C	—	×	×	—	○	○	○	○	×				
30	SP80V-MU	KXF-1K3C	—	×	×	—	○	○	○	○	×				
31	SP80V-L	KXF-1H4C	—	×	×	—	○	○	○	○	×				

(○: Available, ×: Unsupported, —: Not applicable)

*1 Please make an inquiry about any other machines than those of the described above, separately.
*2 NPM X/NPM/AM series, SPV, SPD, SPG and SP70 (LNB connection) support Production Changeover & Control by the machine side options.
*3 The marks at the top in the SP70 fields show the interface conditions that are available when it is connected to LNB, and, the marks at the bottom in the fields, the interface conditions that are available when it is connected to PT200.
*4 If using a mixed line of NPM X/NPM/AM series and CM series (CM602, CM101, CM232, CM212, CM402/ DT401), please contact us.

Machines for Maintenance modules

No.	Model ID	Model No.	Maintenance: Asset profile management	Maintenance: Maintenance history management	Maintenance: Spare parts replacement history management	Maintenance: Prevention of misuse of bad asset	Maintenance: Maintenance information centralized monitor*1
1	NPM-WXS	NM-EJM2E	○	○	○	○	○
2	NPM-WX	NM-EJM9D	○	○	○	○	○
3	NPM-DX	NM-EJM8D	○	○	○	○	○
4	NPM-D3	NM-EJM6D	○	○	○	○	○
5	NPM-D2	NM-EJM5D	○	○	○	○	○
6	NPM-D	NM-EJM1D	○	○	○	○	○
7	NPM-TT2	NM-EJM1E	○	○	○	○	○
8	NPM-TT	NM-EJM3D	○	○	○	○	○
9	NPM-W2S	NM-EJM5D	○	○	○	○	○
10	NPM-W2	NM-EJM7D	○	○	○	○	○
11	NPM-W	NM-EJM2D	○	○	○	○	○
12	NPM	NM-EJM9B	○	○	○	○	○
13	AM100	NM-EJM4D	○	○	○	○	○
14	CM602-L	NM-EJM8A	×	×	×	×	×
15	CM232-M	NM-EJM9C	×	×	×	×	×
16	CM212-M	NM-EJM6A	×	×	×	×	×
17	CM101-D	NM-EJM5B	×	×	×	×	×
18	CM402-L	KXF-4Z4C	×	×	×	×	×
19	CM402-M	KXF-4Y3C	×	×	×	×	×
20	CM401-L	NM-EJM1A	×	×	×	×	×
21	CM401-M	KXF-4X3C	×	×	×	×	×
22	DT401-F	KXF-E64C	×	×	×	×	×
23	DT401-M	KXF-E53C	×	×	×	×	×

(○: Available, ×: Unsupported, —: Not applicable)

*1 Production Monitor & Dispatch module license is required.

Machines for Maintenance modules

No.	Model ID	Model No.	Maintenance: Asset profile management	Maintenance: Maintenance history management	Maintenance: Spare parts replacement history management	Maintenance: Prevention of misuse of bad asset	Maintenance: Maintenance information centralized monitor*1
24	SPV	NM-EJP7A NM-EJP8A	×	×	×	×	×
25	SPD	NM-EJP5A	×	×	×	×	×
26	SPG	NM-EJP6A	×	×	×	×	×
27	SP70※3	NM-EJP3A	×	×	×	×	×
			×	×	×	×	×
28	SP60P-MU	KXF-1J3C	×	×	×	×	×
29	SP60P-L	KXF-1G4C	×	×	×	×	×
30	SP80V-MU	KXF-1K3C	×	×	×	×	×
31	SP80V-L	KXF-1H4C	×	×	×	×	×

(○: Available, ×: Unsupported, －: Not applicable)

*1 Production Monitor & Dispatch module license is required.

Machines for Enterprise Link

Enterprise Link can export data collected by the following modules: Material verification module, Traceability module, and Production analysis module. Machines for Enterprise Link depend on a machine available for those modules; Material verification module, Traceability module, Production analysis module.

5.2 About Regulations

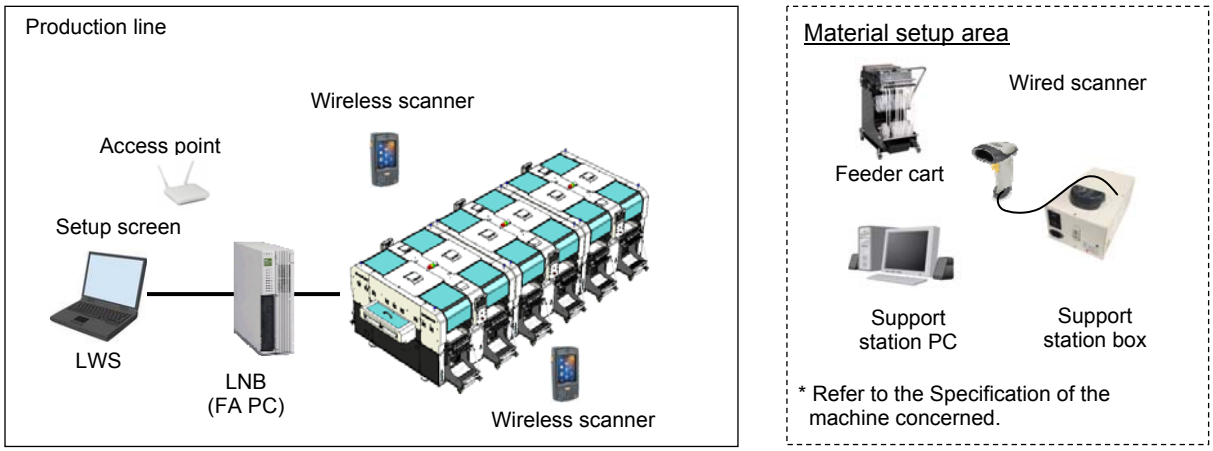
Windows Server® 2008, Windows Server® 2012, Windows Server® 2016, Windows® 7, Windows® 8.1,Microsoft® SQL Server® 2008, Microsoft® SQL Server® 2012, Microsoft® SQL Server® 2016, and Red Hat Linux are export-controlled items (no permission required). Given that MES for Electronics Assembly is application software, you will basically have hardware and OS ready.

6. Serverless Verification Package

From the NPM X series, the LNB parts verification option has been incorporated in the product system of the PanaCIM-EE Material Verification. Thus, PanaCIM-EE Material Verification (serverless type) is provided as a new package.

Selecting this package allows you to build up a system with wireless scanners, LNB and machines that can verify parts line by line without having to install the PanaCIM server, similarly to the conventional LNB parts verification option.

■ System configuration



■ List of functions

Item		Description
Feeder verification※1	Verification at setup	Checks parts at setup.
	Verification at parts exhaust	Checks parts after they have been replenished after part exhaustion.
	Verification at splicing	Checks parts at splicing.
Periodical check		Checks parts at shift change, etc. (at desired timing). It is not necessary to stop production during periodical check.
Recheck	Recheck after feeder jam reset	Rechecks parts after a feeder jam has been reset.
	Recheck after splicing misdetection reset	Rechecks parts after a misdetected splicing has been reset.
	Feeder removal/insert recheck	Rechecks parts if feeder has been removed/inserted due to some reason during a single stop.
Line selection		Lines can be selected on the wireless scanner (PDA) screen. Wireless scanners (PDA) for other lines may also be used.
Setting functions	Verification parts information definition	Parts information to be read from barcode at verification can be defined according to the customer's needs. By including the codes indicating the part class, it is no longer necessary to define the order of scanning.
	Barcode setting	Used to define barcodes. Complicated multi-barcodes are also supported. (Even a barcode containing several meanings can be read by a single scan.)

*1 Stick feeder parts are not supported.

■ Basic specification

Item	Description
Readable codes	 1D code (barcode): UPC/EAN/JAN, UCC/EAN 128, Code 39, Code 128, etc.
	 2D code (2-dimensional code): Maxicode, Data Matrix (ECC 200), QR code, etc.
Code restrictions	Restrictions on the use of codes for part names, lot names and vendor names, etc. Up to 30 ASCII characters and some symbols (- + = , . _ @).
Language	English, Japanese, Chinses*1 1) Displayed on the LWS setting terminal 2) Displayed on wireless scanner (PDA)

*1 Chinese display on the wireless scanner (PDA) is currently under development.

■ Basic configuration

Item	Description	
Software	The wireless scanner (PDA) software is included in the machine/LNB system software DVD-ROM. the same as the conventional LNB parts verification option. ・Machine/LNB system software DVD-ROM ・Operating Instructions DVD-ROM*1 * PanaCIM-EE system software DVD-ROM, PanaCIM license DVD-ROM, and USB dongle are not included in this package.	
Wireless scanner	Refer to 3.3 Hardware/Software Specifications.	
Access point		
License	PanaCIM-EE	(License of PanaCIM-EE Material Verification itself is not required.)
	NPM X/NPM/AM series	PanaCIM-EE Material Verification Serverless type (License)※2
	Screen printers (supported by LNB)	PanaCIM-EE Material Verification Serverless type (License)※2

*1 Instead of existing PanaCIM document, the manual for “PanaCIM-EE Material Verification (Serverless type)” is provided.
*2 Additional machine option for “Serverless Verification Package”. This option includes the ”Material Verification (License)”.

■ Differences from PanaCIM-EE Material Verification (Basic Package)

Item	Description
Applicable models	Applicable only to the machines supported by LNB. CM series and some models are not supported. ・NPM X/NPM/AM series ・Screen printers (SP70, SPD, SPG or later models)
Basic functions	Based on the conventional LNB parts verification option functions. The wireless scanner application directly communicates with the LNB, similarly to the LNB parts verification option, which enables verification without PanaCIM server.
Wireless scanner application	Based on the PanaCIM wireless scanner application. Available functions are limited to use on a line-by-line basis.
Verification setting	LWS screen on the LNB is used. Setting required for each line.
Off-line verification	The following are required to perform off-line verification: ・Support station “Parts collation type (license)” ・Wired scanner and support station box/PC (Wireless scanner not supported)
Server management functions	Since no server is included, the following functions are not supported ・Material control via reel ID (quantities, location, moisture-sensitive devices, etc.) ・Functions across several lines (Line changeover scan, duplicate reel ID check, etc.)

7. Regarding License

Differences in software license between existing software, such as PanaPRO/IPO, and PanaCIM-EE are described below.

Item	Existing software (PanaPRO/IPO, etc.)	PanaCIM-EE
License type	Limited license	Time-limited (one year) license * Including a 1-year upgrade contract
	Permanent license	Permanent license * Upgrade contract is needed, separately. (option)
License units	Other than Maintenance module	Number of machine stages e.g. CM602-L: 2 licenses NPM-D3: 1 license
	Maintenance module	Number of machine stages * Up to 500 items can be registered per stage. (Assets such as heads, feeders, and nozzles that are subject to maintenance are referred to as items.)
Upgrade agreement		Upgrade DVD is provided to you for an annual upgrade contract. (The upgrade fee is required, separately)

- A license is required per PanaCIM-EE module.
  (A license is required for each of the functions (“Basic”, “Material Allocation” and “Moisture Control”) for the material control module).
- When using the Component Supply Navigator (an option of Production Monitoring & Dispatch Module), the Production Monitoring & Dispatch Module (license) and Material verification module (license) are required.
- When using the Maintenance information centralized monitor of the Maintenance module, the Production Monitoring & Dispatch Module license is required.
- License is offered in the form of license DVD. Customer is responsible for storing it.

Terms and conditions regarding the use of software

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   (4) Any defects caused by accident, misuse, neglect, abuse, mishandling, misapplication, modification, alternation, acts of God, or improper installation or maintenance; or
   (5) Any defects arising out of modification or attempted modification of the Software by you.

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Important Notes

1. Performance assurance of the Product is subject to the condition that the Product is used in accordance with specifications, specification manuals, and operating instructions provided by us. We do not guarantee performance if the Product is used in any way other than those described in such documents.
   (a) Never use the Product near any noise generation sources such as welding machines, which will result in malfunctions.
   (b) Do not unnecessarily turn off the power switch, servo switch, or air during operation; it will run without being controlled, which will result in damage.
   (c) Before teaching operations or conducting manual operations in a servo free state, make sure there is no interference in the unit, which will result in damage.
   (d) The covers of the main body, loader/unloader, tray feeder, and the other covers may be cylinder-locked.
   Do not attempt to forcibly open the covers, which will result in damage to the covers and/or cylinders.
   (e) Do not take apart or alter the equipment and its units, as it may influence product safety and/or its quality.
   (f) Data modifications must only be undertaken by professionally trained and approved personnel. Modifications by unqualified personnel may be a cause of input error, which will result in malfunctions.
   (g) Do not block any of exhaust openings and intakes. Excessive temperature in the control box is a cause of abnormal functioning.
   (h) Do not turn off the servo switch during operation, as it may result in control failure.
   Always set the Product to a temporarily paused state (e.g. single stop or cycle stop) or operation stop state before the Product is turned off.
   (i) Be attentive to water quality when using spraying devices and/or humidifiers. In an environment where ultrasonic humidifiers are used, for example, mineral components such as calcium or magnesium, and components such as silica, carbonate, or iron, are released to the air. Such airborne particles stick to the equipment, resulting in the malfunction of sliding parts and/or optical components.
   (j) Do not use the Product in an atmosphere where gasified low molecular weight siloxane exists.
   Low molecular siloxane, released by exposed silicone-based materials (e.g. silicone adhesives, silicone oils, or silicone powders), sticks to the electric contact parts and/or optical components, and will result in contact failure and/or malfunction.
   (k) Do not use the Product in corrosive gas environments (e.g. nitric acid, hydrogen sulfide, or ammonia), halogen gas environments, or any environment where conductive foreign substances may enter.
   Electrodes and/or metal parts of electric components will corrode, and will become a cause of the malfunction of electronic circuits and/or malfunctions.

2. We shall not be held liable for the following failures and damages.
   (a) Failure and/or damage caused by wrong use or improper repair, adjustment or modification
   (b) Incidental damage (e.g. loss of profits or the interruption of business) arising from the use of or inability to use the Product
   (c) Failure and/or damage caused by relocation or transportation of the Product, the dropping of the Product itself or by objects falling on the Product after purchase
   (d) Failure and/or damage caused by fire, earthquake, flood, thunderbolt, other natural or man-made disasters, pollution damage, salt damage, gas damage (e.g. corrosive gas or halogen gas), abnormal voltage, or the use of power sources besides those specified (e.g. voltage or frequency)
   (e) Failure and/or damage caused by carriers during delivery via vehicle, ship, or other modes of transport.
   (f) Damage arising from failure to observe the instructions/cautions of the operating instructions provided with the Product.
   (g) Any failure caused by hacking, unauthorized access, attacks upon the Product or connected networks and from other similar threats

3. (a) We shall not be held liable for any damage caused by any dispute alleging or claiming the infringement of intellectual property rights and other rights arising in connection with manufacturing inspections or other processes performed during the use of the Products, including any allegation or claim that any items manufactured and/or inspected using the Products is infringing intellectual property rights or other rights of any third party.
   (b) We shall not be held liable for any allegation or claim of infringement which has been settled or which judgment has been reached without our express written consent.

4. We shall in no event be held liable for any indirect or consequential damages, or for loss of profit that may arise in connection with the Products or their use.

5. The user shall be responsible for determining the necessary structural reinforcement or modifications to the building in which the Products will be installed and for the completion of such measures, all at its own cost and expense, prior to the date of delivery of the Products.

Some of our products fall under the export control items (or technologies) established in the Foreign Exchange and Foreign Trade Act. When exporting (or providing technologies overseas), please check its legal restrictions and be sure to take necessary procedures such as to obtain the export permission.

Panasonic Smart Factory Solutions Co., Ltd.

1375 Kamisukiawara, Showa-cho, Nakakoma-gun, Yamanashi 409-3895, Japan

* Recycled paper is used for this Specification.